

In general it may not be possible to lift a homomorphism f from D to N to a homomorphism from D to M . For example, consider the nonsplit exact sequence $0 \rightarrow \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/2\mathbb{Z} \rightarrow 0$ from the previous set of examples. Let $D = \mathbb{Z}/2\mathbb{Z}$ and let f be the identity map from D into N . Any homomorphism F of D into $M = \mathbb{Z}$ must map D to 0 (since $\mathbb{Z}$ has no elements of order 2), hence $\pi \circ F$ maps D to 0 in N , and in particular, $\pi \circ F \neq f$. Phrased in terms of the map φ' , this shows that

$$\text{if } M \xrightarrow{\varphi} N \rightarrow 0 \text{ is exact,}$$

$$\text{then } \text{Hom}_R(D, M) \xrightarrow{\varphi'} \text{Hom}_R(D, N) \rightarrow 0 \text{ is not necessarily exact.}$$

These results relating the homomorphisms into L and N to the homomorphisms into M can be neatly summarized as part of the following theorem.

Theorem 28. Let D , L , M , and N be R -modules. If

$$0 \rightarrow L \xrightarrow{\psi} M \xrightarrow{\varphi} N \rightarrow 0 \text{ is exact,}$$

then the associated sequence

$$0 \rightarrow \text{Hom}_R(D, L) \xrightarrow{\psi'} \text{Hom}_R(D, M) \xrightarrow{\varphi'} \text{Hom}_R(D, N) \text{ is exact.} \quad (10.10)$$

A homomorphism $f : D \rightarrow N$ lifts to a homomorphism $F : D \rightarrow M$ if and only if $f \in \text{Hom}_R(D, N)$ is in the image of φ' . In general $\varphi' : \text{Hom}_R(D, M) \rightarrow \text{Hom}_R(D, N)$ need not be surjective; the map φ' is surjective if and only if every homomorphism from D to N lifts to a homomorphism from D to M , in which case the sequence (10) can be extended to a short exact sequence.

The sequence (10) is exact for *all* R -modules D if and only if the sequence

$$0 \rightarrow L \xrightarrow{\psi} M \xrightarrow{\varphi} N \text{ is exact.}$$

Proof: The only item in the first statement that has not already been proved is the exactness of (10) at $\text{Hom}_R(D, M)$, i.e., $\ker \varphi' = \text{image } \psi'$. Suppose $F : D \rightarrow M$ is an element of $\text{Hom}_R(D, M)$ lying in the kernel of φ' , i.e., with $\varphi \circ F = 0$ as homomorphisms from D to N . If $d \in D$ is any element of D , this implies that $\varphi(F(d)) = 0$ and $F(d) \in \ker \varphi$. By the exactness of the sequence defining the extension M we have $\ker \varphi = \text{image } \psi$, so there is some element $l \in L$ with $F(d) = \psi(l)$. Since ψ is injective, the element l is unique, so this gives a well defined map $F' : D \rightarrow L$ given by $F'(d) = l$. It is an easy check to verify that F' is a homomorphism, i.e., $F' \in \text{Hom}_R(D, L)$. Since $\psi \circ F'(d) = \psi(l) = F(d)$, we have $F = \psi'(F')$ which shows that F is in the image of ψ' , proving that $\ker \varphi' \subseteq \text{image } \psi'$. Conversely, if F is in the image of ψ' then $F = \psi'(F')$ for some $F' \in \text{Hom}_R(D, L)$ and so $\varphi(F(d)) = \varphi(\psi(F'(d)))$ for any $d \in D$. Since $\ker \varphi = \text{image } \psi$ we have $\varphi \circ \psi = 0$, and it follows that $\varphi(F(d)) = 0$ for any $d \in D$, i.e., $\varphi'(F) = 0$. Hence F is in the kernel of φ' , proving the reverse containment: $\text{image } \psi' \subseteq \ker \varphi'$.

For the last statement in the theorem, note first that the surjectivity of φ was not required for the proof that (10) is exact, so the “if” portion of the statement has already

been proved. For the converse, suppose that the sequence (10) is exact for all R -modules D . In general, $\text{Hom}_R(R, X) \cong X$ for any left R -module X , the isomorphism being given by mapping a homomorphism to its value on the element $1 \in R$ (cf. Exercise 10(b)). Taking $D = R$ in (10), the exactness of the sequence $0 \rightarrow L \xrightarrow{\psi} M \xrightarrow{\varphi} N$ follows easily.

By Theorem 28, the sequence

$$0 \rightarrow \text{Hom}_R(D, L) \xrightarrow{\psi'} \text{Hom}_R(D, M) \xrightarrow{\varphi'} \text{Hom}_R(D, N) \rightarrow 0 \quad (10.11)$$

is in general *not* a short exact sequence since the homomorphism φ' need not be surjective. The question of whether this sequence is exact precisely measures the extent to which the homomorphisms from D into M are uniquely determined by pairs of homomorphisms from D into L and D into N . More precisely, this sequence is exact if and only if there is a bijection $F \leftrightarrow (g, f)$ between homomorphisms $F : D \rightarrow M$ and pairs of homomorphisms $g : D \rightarrow L$ and $f : D \rightarrow N$ given by $F|_{\psi(L)} = \psi'(g)$ and $f = \varphi'(F)$.

One situation in which the sequence (11) is exact occurs when the original sequence $0 \rightarrow L \rightarrow M \rightarrow N \rightarrow 0$ is a *split* exact sequence, i.e., when $M = L \oplus N$. In this case the sequence (11) is also a split exact sequence, as the first part of the following proposition shows.

Proposition 29. Let D, L and N be R -modules. Then

- (1) $\text{Hom}_R(D, L \oplus N) \cong \text{Hom}_R(D, L) \oplus \text{Hom}_R(D, N)$, and
- (2) $\text{Hom}_R(L \oplus N, D) \cong \text{Hom}_R(L, D) \oplus \text{Hom}_R(N, D)$.

Proof: Let $\pi_1 : L \oplus N \rightarrow L$ be the natural projection from $L \oplus N$ to L and similarly let π_2 be the natural projection to N . If $f \in \text{Hom}_R(D, L \oplus N)$ then the compositions $\pi_1 \circ f$ and $\pi_2 \circ f$ give elements in $\text{Hom}_R(D, L)$ and $\text{Hom}_R(D, N)$, respectively. This defines a map from $\text{Hom}_R(D, L \oplus N)$ to $\text{Hom}_R(D, L) \oplus \text{Hom}_R(D, N)$ which is easily seen to be a homomorphism. Conversely, given $f_1 \in \text{Hom}_R(D, L)$ and $f_2 \in \text{Hom}_R(D, N)$, define the map $f \in \text{Hom}_R(D, L \oplus N)$ by $f(d) = (f_1(d), f_2(d))$. This defines a map from $\text{Hom}_R(D, L) \oplus \text{Hom}_R(D, N)$ to $\text{Hom}_R(D, L \oplus N)$ that is easily checked to be a homomorphism inverse to the map above, proving the isomorphism in (1). The proof of (2) is similar and is left as an exercise.

The results in Proposition 29 extend immediately by induction to any finite direct sum of R -modules. These results are referred to by saying that *Hom commutes with finite direct sums in either variable* (compare to Theorem 17 for a corresponding result for tensor products). For infinite direct sums the situation is more complicated. Part (1) remains true if $L \oplus N$ is replaced by an arbitrary direct sum and the direct sum on the right hand side is replaced by a direct product (Exercise 13 shows that the direct product is necessary). Part (2) remains true if the direct sums on both sides are replaced by direct products.

This proposition shows that if the sequence

$$0 \rightarrow L \xrightarrow{\psi} M \xrightarrow{\varphi} N \rightarrow 0$$

is a split short exact sequence of R -modules, then

$$0 \longrightarrow \operatorname{Hom}_R(D, L) \xrightarrow{\psi'} \operatorname{Hom}_R(D, M) \xrightarrow{\varphi'} \operatorname{Hom}_R(D, N) \longrightarrow 0$$

is also a split short exact sequence of abelian groups for every R -module D . Exercise 14 shows that a converse holds: if $0 \rightarrow \operatorname{Hom}_R(D, L) \xrightarrow{\psi'} \operatorname{Hom}_R(D, M) \xrightarrow{\varphi'} \operatorname{Hom}_R(D, N) \rightarrow 0$ is exact for every R -module D then $0 \rightarrow L \xrightarrow{\psi} M \xrightarrow{\varphi} N \rightarrow 0$ is a split short exact sequence (which then implies that if the original Hom sequence is exact for every D , then in fact it is split exact for every D).

Proposition 29 identifies a situation in which the sequence (11) is exact in terms of the modules L , M , and N . The next result adopts a slightly different perspective, characterizing instead the modules D having the property that the sequence (10) in Theorem 28 can *always* be extended to a short exact sequence:

Proposition 30. Let P be an R -module. Then the following are equivalent:

- (1) For any R -modules L , M , and N , if

$$0 \longrightarrow L \xrightarrow{\psi} M \xrightarrow{\varphi} N \longrightarrow 0$$

is a short exact sequence, then

$$0 \longrightarrow \operatorname{Hom}_R(P, L) \xrightarrow{\psi'} \operatorname{Hom}_R(P, M) \xrightarrow{\varphi'} \operatorname{Hom}_R(P, N) \longrightarrow 0$$

is also a short exact sequence.

- (2) For any R -modules M and N , if $M \xrightarrow{\varphi} N \rightarrow 0$ is exact, then every R -module homomorphism from P into N lifts to an R -module homomorphism into M , i.e., given $f \in \operatorname{Hom}_R(P, N)$ there is a lift $F \in \operatorname{Hom}_R(P, M)$ making the following diagram commute:

$$\begin{array}{ccc} & P & \\ \swarrow F & \downarrow f & \\ M & \xrightarrow{\varphi} & N \longrightarrow 0 \end{array}$$

- (3) If P is a quotient of the R -module M then P is isomorphic to a direct summand of M , i.e., every short exact sequence $0 \rightarrow L \rightarrow M \rightarrow P \rightarrow 0$ splits.
 (4) P is a direct summand of a free R -module.

Proof: The equivalence of (1) and (2) is a restatement of a result in Theorem 28.

Suppose now that (2) is satisfied, and let $0 \rightarrow L \xrightarrow{\psi} M \xrightarrow{\varphi} P \rightarrow 0$ be exact. By (2), the identity map from P to P lifts to a homomorphism μ making the following diagram commute:

$$\begin{array}{ccc} & P & \\ \swarrow \mu & \downarrow id & \\ M & \xrightarrow{\varphi} & P \longrightarrow 0 \end{array}$$

Then $\varphi \circ \mu = 1$, so μ is a splitting homomorphism for the sequence, which proves (3).

Every module P is the quotient of a free module (for example, the free module on the

set of elements in P), so there is always an exact sequence $0 \rightarrow \ker \varphi \rightarrow \mathcal{F} \xrightarrow{\varphi} P \rightarrow 0$ where $\mathcal{F}$ is a free R -module (cf. Example 4 following Corollary 23). If (3) is satisfied, then this sequence splits, so $\mathcal{F}$ is isomorphic to the direct sum of $\ker \varphi$ and P , which proves (4).

Finally, to prove (4) implies (2), suppose that P is a direct summand of a free R -module on some set S , say $\mathcal{F}(S) = P \oplus K$, and that we are given a homomorphism f from P to N as in (2). Let π denote the natural projection from $\mathcal{F}(S)$ to P , so that $f \circ \pi$ is a homomorphism from $\mathcal{F}(S)$ to N . For any $s \in S$ define $n_s = f \circ \pi(s) \in N$ and let $m_s \in M$ be any element of M with $\varphi(m_s) = n_s$ (which exists because φ is surjective). By the universal property for free modules (Theorem 6 of Section 3), there is a unique R -module homomorphism F' from $\mathcal{F}(S)$ to M with $F'(s) = m_s$. The diagram is the following:

$$\begin{array}{ccc} & \mathcal{F}(S) = P \oplus K & \\ & \downarrow \pi & \\ & P & \\ & \downarrow f & \\ M & \xrightarrow{\varphi} & N \longrightarrow 0 \end{array}$$

(Note: A dashed arrow labeled F' points from $\mathcal{F}(S)$ to M .)

By definition of the homomorphism F' we have $\varphi \circ F'(s) = \varphi(m_s) = n_s = f \circ \pi(s)$, from which it follows that $\varphi \circ F' = f \circ \pi$ on $\mathcal{F}(S)$, i.e., the diagram above is commutative. Now define a map $F : P \rightarrow M$ by $F(d) = F'((d, 0))$. Since F is the composite of the injection $P \rightarrow \mathcal{F}(S)$ with the homomorphism F' , it follows that F is an R -module homomorphism. Then

$$\varphi \circ F(d) = \varphi \circ F'((d, 0)) = f \circ \pi((d, 0)) = f(d)$$

i.e., $\varphi \circ F = f$, so the diagram

$$\begin{array}{ccc} & P & \\ & \downarrow f & \\ M & \xrightarrow{\varphi} & N \longrightarrow 0 \end{array}$$

(Note: A dashed arrow labeled F points from P to M .)

commutes, which proves that (4) implies (2) and completes the proof.

Definition. An R -module P is called *projective* if it satisfies any of the equivalent conditions of Proposition 30.

The third statement in Proposition 30 can be rephrased as saying that any module M that projects onto P has (an isomorphic copy of) P as a direct summand, which explains the terminology.

The following result is immediate from Proposition 30 (and its proof):

Corollary 31. Free modules are projective. A finitely generated module is projective if and only if it is a direct summand of a finitely generated free module. Every module is a quotient of a projective module.

If D is fixed, then given any R -module X we have an associated abelian group $\text{Hom}_R(D, X)$. Further, an R -module homomorphism $\alpha : X \rightarrow Y$ induces an abelian group homomorphism $\alpha' : \text{Hom}_R(D, X) \rightarrow \text{Hom}_R(D, Y)$, defined by $\alpha'(f) = \alpha \circ f$. Put another way, the map $\text{Hom}_R(D, _)$ is a *covariant functor* from the category of R -modules to the category of abelian groups (cf. Appendix II). Theorem 28 shows that applying this functor to the terms in the exact sequence

$$0 \longrightarrow L \xrightarrow{\psi} M \xrightarrow{\varphi} N \longrightarrow 0$$

produces an exact sequence

$$0 \rightarrow \text{Hom}_R(D, L) \xrightarrow{\psi'} \text{Hom}_R(D, M) \xrightarrow{\varphi'} \text{Hom}_R(D, N).$$

This is referred to by saying that $\text{Hom}_R(D, _)$ is a *left exact* functor. By Proposition 30, the functor $\text{Hom}_R(D, _)$ is *exact*, i.e., always takes short exact sequences to short exact sequences, if and only if D is projective. We summarize this as

Corollary 32. If D is an R -module, then the functor $\text{Hom}_R(D, _)$ from the category of R -modules to the category of abelian groups is left exact. It is exact if and only if D is a projective R -module.

Note that if $\text{Hom}_R(D, _)$ takes short exact sequences to short exact sequences, then it takes exact sequences of any length to exact sequences since any exact sequence can be broken up into a succession of short exact sequences.

As we have seen, the functor $\text{Hom}_R(D, _)$ is in general not exact on the right. Measuring the extent to which functors such as $\text{Hom}_R(D, _)$ fail to be exact leads to the notions of “homological algebra,” considered in Chapter 17.

Examples

- (1) We shall see in Section 11.1 that if $R = F$ is a field then every F -module is projective (although we only prove this for finitely generated modules).
- (2) By Corollary 31, $\mathbb{Z}$ is a projective $\mathbb{Z}$ -module. This can be seen directly as follows: suppose f is a map from $\mathbb{Z}$ to N and $M \xrightarrow{\varphi} N \rightarrow 0$ is exact. The homomorphism f is uniquely determined by the value $n = f(1)$. Then f can be lifted to a homomorphism $F : \mathbb{Z} \rightarrow M$ by first defining $F(1) = m$, where m is any element in M mapped to n by φ , and then extending F to all of $\mathbb{Z}$ by additivity.

By the first statement in Proposition 30, since $\mathbb{Z}$ is projective, if

$$0 \longrightarrow L \xrightarrow{\psi} M \xrightarrow{\varphi} N \longrightarrow 0$$

is an exact sequence of $\mathbb{Z}$ -modules, then

$$0 \longrightarrow \text{Hom}_{\mathbb{Z}}(\mathbb{Z}, L) \xrightarrow{\psi'} \text{Hom}_{\mathbb{Z}}(\mathbb{Z}, M) \xrightarrow{\varphi'} \text{Hom}_{\mathbb{Z}}(\mathbb{Z}, N) \longrightarrow 0$$

is also an exact sequence. This can also be seen directly using the isomorphism $\text{Hom}_{\mathbb{Z}}(\mathbb{Z}, M) \cong M$ of abelian groups, which shows that the two exact sequences above are essentially the same.

- (3) Free $\mathbb{Z}$ -modules have no nonzero elements of finite order so no nonzero finite abelian group can be isomorphic to a submodule of a free module. By Corollary 31 it follows that no nonzero finite abelian group is a projective $\mathbb{Z}$ -module.

- (4) As a particular case of the preceding example, we see that for $n \geq 2$ the $\mathbb{Z}$ -module $\mathbb{Z}/n\mathbb{Z}$ is not projective. By Theorem 28 it must be possible to find a short exact sequence which after applying the functor $\text{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, _)$ is no longer exact on the right. One such sequence is the exact sequence of Example 2 following Corollary 23:

$$0 \longrightarrow \mathbb{Z} \xrightarrow{n} \mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/n\mathbb{Z} \longrightarrow 0,$$

for $n \geq 2$. Note first that $\text{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}) = 0$ since there are no nonzero $\mathbb{Z}$ -module homomorphisms from $\mathbb{Z}/n\mathbb{Z}$ to $\mathbb{Z}$. It is also easy to see that $\text{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/n\mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$, as follows. Every homomorphism f is uniquely determined by $f(1) = a \in \mathbb{Z}/n\mathbb{Z}$, and given any $a \in \mathbb{Z}/n\mathbb{Z}$ there is a unique homomorphism f_a with $f_a(1) = a$; the map $f_a \mapsto a$ is easily checked to be an isomorphism from $\text{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/n\mathbb{Z})$ to $\mathbb{Z}/n\mathbb{Z}$.

Applying $\text{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, _)$ to the short exact sequence above thus gives the sequence

$$0 \longrightarrow 0 \xrightarrow{n'} 0 \xrightarrow{\pi'} \mathbb{Z}/n\mathbb{Z} \longrightarrow 0$$

which is not exact at its only nonzero term.

- (5) Since $\mathbb{Q}/\mathbb{Z}$ is a torsion $\mathbb{Z}$ -module it is not a submodule of a free $\mathbb{Z}$ -module, hence is not projective. Note also that the exact sequence $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \xrightarrow{\pi} \mathbb{Q}/\mathbb{Z} \rightarrow 0$ does not split since $\mathbb{Q}$ contains no submodule isomorphic to $\mathbb{Q}/\mathbb{Z}$.
- (6) The $\mathbb{Z}$ -module $\mathbb{Q}$ is not projective (cf. the exercises).
- (7) We shall see in Chapter 12 that a finitely generated $\mathbb{Z}$ -module is projective if and only if it is free.
- (8) Let R be the commutative ring $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ under componentwise addition and multiplication. If P_1 and P_2 are the principal ideals generated by $(1, 0)$ and $(0, 1)$ respectively then $R = P_1 \oplus P_2$, hence both P_1 and P_2 are projective R -modules by Proposition 30. Neither P_1 nor P_2 is free, since any free module has order a multiple of four.
- (9) The direct sum of two projective modules is again projective (cf. Exercise 3).
- (10) We shall see in Part VI that if F is any field and $n \in \mathbb{Z}^+$ then the ring $R = M_n(F)$ of all $n \times n$ matrices with entries from F has the property that every R -module is projective. We shall also see that if G is a finite group of order n and $n \neq 0$ in the field F then the group ring FG also has the property that every module is projective.

Injective Modules and $\text{Hom}_R(_, D)$

If $0 \longrightarrow L \xrightarrow{\psi} M \xrightarrow{\varphi} N \longrightarrow 0$ is a short exact sequence of R -modules then, instead of considering maps from an R -module D into L or N and the extent to which these determine maps from D into M , we can consider the “dual” question of maps from L or N to D . In this case, it is easy to dispose of the situation of a map from N to D : an R -module map from N to D immediately gives a map from M to D simply by composing with φ . It is easy to check that this defines an injective homomorphism of abelian groups

$$\begin{aligned} \varphi' : \text{Hom}_R(N, D) &\longrightarrow \text{Hom}_R(M, D) \\ f &\longmapsto f' = f \circ \varphi, \end{aligned}$$

or, put another way,

$$\text{if } M \xrightarrow{\varphi} N \rightarrow 0 \text{ is exact,}$$

$$\text{then } 0 \rightarrow \text{Hom}_R(N, D) \xrightarrow{\varphi'} \text{Hom}_R(M, D) \text{ is exact.}$$

(Note that the associated maps on the homomorphism groups are in the reverse direction from the original maps.)

On the other hand, given an R -module homomorphism f from L to D it may not be possible to extend f to a map F from M to D , i.e., given f it may not be possible to find a map F making the following diagram commute:

$$\begin{array}{ccc} L & \xrightarrow{\psi} & M \\ f \downarrow & \swarrow F & \\ D & & \end{array}$$

For example, consider the exact sequence $0 \rightarrow \mathbb{Z} \xrightarrow{\psi} \mathbb{Z} \xrightarrow{\varphi} \mathbb{Z}/2\mathbb{Z} \rightarrow 0$ of $\mathbb{Z}$ -modules, where ψ is multiplication by 2 and φ is the natural projection. Take $D = \mathbb{Z}/2\mathbb{Z}$ and let $f : \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$ be reduction modulo 2 on the first $\mathbb{Z}$ in the sequence. There is only one nonzero homomorphism F from the second $\mathbb{Z}$ in the sequence to $\mathbb{Z}/2\mathbb{Z}$ (namely, reduction modulo 2), but this F does not lift the map f since $F \circ \psi(\mathbb{Z}) = F(2\mathbb{Z}) = 0$, so $F \circ \psi \neq f$.

Composition with ψ induces an abelian group homomorphism ψ' from $\text{Hom}_R(M, D)$ to $\text{Hom}_R(L, D)$, and in terms of the map ψ' , the homomorphism $f \in \text{Hom}_R(L, D)$ can be lifted to a homomorphism from M to D if and only if f is in the image of ψ' . The example above shows that

$$\text{if } 0 \rightarrow L \xrightarrow{\psi} M \text{ is exact,}$$

$$\text{then } \text{Hom}_R(M, D) \xrightarrow{\psi'} \text{Hom}_R(L, D) \rightarrow 0 \text{ is not necessarily exact.}$$

We can summarize these results in the following dual version of Theorem 28:

Theorem 33. Let D, L, M , and N be R -modules. If

$$0 \rightarrow L \xrightarrow{\psi} M \xrightarrow{\varphi} N \rightarrow 0 \text{ is exact,}$$

then the associated sequence

$$0 \rightarrow \text{Hom}_R(N, D) \xrightarrow{\varphi'} \text{Hom}_R(M, D) \xrightarrow{\psi'} \text{Hom}_R(L, D) \text{ is exact.} \quad (10.12)$$

A homomorphism $f : L \rightarrow D$ lifts to a homomorphism $F : M \rightarrow D$ if and only if $f \in \text{Hom}_R(L, D)$ is in the image of ψ' . In general $\psi' : \text{Hom}_R(M, D) \rightarrow \text{Hom}_R(L, D)$ need not be surjective; the map ψ' is surjective if and only if every homomorphism from L to D lifts to a homomorphism from M to D , in which case the sequence (12) can be extended to a short exact sequence.

The sequence (12) is exact for *all* R -modules D if and only if the sequence

$$L \xrightarrow{\psi} M \xrightarrow{\varphi} N \rightarrow 0 \text{ is exact.}$$

Proof: The only item remaining to be proved in the first statement is the exactness of (12) at $\text{Hom}_R(M, D)$. The proof of this statement is very similar to the proof of the corresponding result in Theorem 28 and is left as an exercise. Note also that the injectivity of ψ is not required, which proves the “if” portion of the final statement of the theorem.

Suppose now that the sequence (12) is exact for all R -modules D . We first show that $\varphi : M \rightarrow N$ is a surjection. Take $D = N/\varphi(M)$. If $\pi_1 : N \rightarrow N/\varphi(M)$ is the natural projection homomorphism, then $\pi_1 \circ \varphi(M) = 0$ by definition of π_1 . Since $\pi_1 \circ \varphi = \varphi'(\pi_1)$, this means that the element $\pi_1 \in \text{Hom}_R(N, N/\varphi(M))$ is mapped to 0 by φ' . Since φ' is assumed to be injective for all modules D , this means π_1 is the zero map, i.e., $N = \varphi(M)$ and so φ is a surjection. We next show that $\varphi \circ \psi = 0$, which will imply that $\text{image } \psi \subseteq \ker \varphi$. For this we take $D = N$ and observe that the identity map id_N on N is contained in $\text{Hom}_R(N, N)$, hence $\varphi'(id_N) \in \text{Hom}_R(M, N)$. Then the exactness of (12) for $D = N$ implies that $\varphi'(id_N) \in \ker \psi'$, so $\psi'(\varphi'(id_N)) = 0$. Then $id_N \circ \psi \circ \varphi = 0$, i.e., $\psi \circ \varphi = 0$, as claimed. Finally, we show that $\ker \varphi \subseteq \text{image } \psi$. Let $D = M/\psi(L)$ and let $\pi_2 : M \rightarrow M/\psi(L)$ be the natural projection. Then $\psi'(\pi_2) = 0$ since $\pi_2(\psi(L)) = 0$ by definition of π_2 . The exactness of (12) for this D then implies that π_2 is in the image of φ' , say $\pi_2 = \varphi'(f)$ for some homomorphism $f \in \text{Hom}_R(N, M/\psi(L))$, i.e., $\pi_2 = f \circ \varphi$. If $m \in \ker \varphi$ then $\pi_2(m) = f(\varphi(m)) = 0$, which means that $m \in \psi(L)$ since π_2 is just the projection from M into the quotient $M/\psi(L)$. Hence $\ker \varphi \subseteq \text{image } \psi$, completing the proof.

By Theorem 33, the sequence

$$0 \longrightarrow \text{Hom}_R(N, D) \xrightarrow{\varphi'} \text{Hom}_R(M, D) \xrightarrow{\psi'} \text{Hom}_R(L, D) \longrightarrow 0$$

is in general *not* a short exact sequence since ψ' need not be surjective, and the question of whether this sequence is exact precisely measures the extent to which homomorphisms from M to D are uniquely determined by pairs of homomorphisms from L and N to D .

The second statement in Proposition 29 shows that this sequence is exact when the original exact sequence $0 \rightarrow L \rightarrow M \rightarrow N \rightarrow 0$ is a *split* exact sequence. In fact in this case the sequence $0 \rightarrow \text{Hom}_R(N, D) \xrightarrow{\varphi'} \text{Hom}_R(M, D) \xrightarrow{\psi'} \text{Hom}_R(L, D) \rightarrow 0$ is also a split exact sequence of abelian groups for every R -module D . Exercise 14 shows that a converse holds: if $0 \rightarrow \text{Hom}_R(N, D) \xrightarrow{\varphi'} \text{Hom}_R(M, D) \xrightarrow{\psi'} \text{Hom}_R(L, D) \rightarrow 0$ is exact for every R -module D then $0 \rightarrow L \xrightarrow{\psi} M \xrightarrow{\varphi} N \rightarrow 0$ is a split short exact sequence (which then implies that if the Hom sequence is exact for every D , then in fact it is split exact for every D).

There is also a dual version of the first three parts of Proposition 30, which describes the R -modules D having the property that the sequence (12) in Theorem 33 can *always* be extended to a short exact sequence:

Proposition 34. Let Q be an R -module. Then the following are equivalent:

- (1) For any R -modules L, M , and N , if

$$0 \longrightarrow L \xrightarrow{\psi} M \xrightarrow{\varphi} N \longrightarrow 0$$

is a short exact sequence, then

$$0 \longrightarrow \operatorname{Hom}_R(N, Q) \xrightarrow{\varphi'} \operatorname{Hom}_R(M, Q) \xrightarrow{\psi'} \operatorname{Hom}_R(L, Q) \longrightarrow 0$$

is also a short exact sequence.

- (2) For any R -modules L and M , if $0 \rightarrow L \xrightarrow{\psi} M$ is exact, then every R -module homomorphism from L into Q lifts to an R -module homomorphism of M into Q , i.e., given $f \in \operatorname{Hom}_R(L, Q)$ there is a lift $F \in \operatorname{Hom}_R(M, Q)$ making the following diagram commute:

$$\begin{array}{ccccc} 0 & \longrightarrow & L & \xrightarrow{\psi} & M \\ & & \downarrow f & \nearrow F & \\ & & Q & & \end{array}$$

- (3) If Q is a submodule of the R -module M then Q is a direct summand of M , i.e., every short exact sequence $0 \rightarrow Q \rightarrow M \rightarrow N \rightarrow 0$ splits.

Proof: The equivalence of (1) and (2) is part of Theorem 33. Suppose now that (2) is satisfied and let $0 \rightarrow Q \xrightarrow{\psi} M \xrightarrow{\varphi} N \rightarrow 0$ be exact. Taking $L = Q$ and f the identity map from Q to itself, it follows by (2) that there is a homomorphism $F : M \rightarrow Q$ with $F \circ \psi = 1$, so F is a splitting homomorphism for the sequence, which proves (3). The proof that (3) implies (2) is outlined in the exercises.

Definition. An R -module Q is called *injective* if it satisfies any of the equivalent conditions of Proposition 34.

The third statement in Proposition 34 can be rephrased as saying that any module M into which Q injects has (an isomorphic copy of) Q as a direct summand, which explains the terminology.

If D is fixed, then given any R -module X we have an associated abelian group $\operatorname{Hom}_R(X, D)$. Further, an R -module homomorphism $\alpha : X \rightarrow Y$ induces an abelian group homomorphism $\alpha' : \operatorname{Hom}_R(Y, D) \rightarrow \operatorname{Hom}_R(X, D)$, defined by $\alpha'(f) = f \circ \alpha$, that “reverses” the direction of the arrow. Put another way, the map $\operatorname{Hom}_R(D, _)$ is a *contravariant functor* from the category of R -modules to the category of abelian groups (cf. Appendix II). Theorem 33 shows that applying this functor to the terms in the exact sequence

$$0 \longrightarrow L \xrightarrow{\psi} M \xrightarrow{\varphi} N \longrightarrow 0$$

produces an exact sequence

$$0 \rightarrow \operatorname{Hom}_R(N, D) \xrightarrow{\varphi'} \operatorname{Hom}_R(M, D) \xrightarrow{\psi'} \operatorname{Hom}_R(L, D).$$

This is referred to by saying that $\operatorname{Hom}_R(_, D)$ is a *left exact* (contravariant) functor. Note that the functor $\operatorname{Hom}_R(_, D)$ and the functor $\operatorname{Hom}_R(D, _)$ considered earlier

are both left exact; the former reverses the directions of the maps in the original short exact sequence, the latter maintains the directions of the maps.

By Proposition 34, the functor $\text{Hom}_R(_, D)$ is *exact*, i.e., always takes short exact sequences to short exact sequences (and hence exact sequences of any length to exact sequences), if and only if D is injective. We summarize this in the following proposition, which is dual to the covariant result of Corollary 32.

Corollary 35. If D is an R -module, then the functor $\text{Hom}_R(_, D)$ from the category of R -modules to the category of abelian groups is left exact. It is exact if and only if D is an injective R -module.

We have seen that an R -module is projective if and only if it is a direct summand of a free R -module. Providing such a simple characterization of injective R -modules is not so easy. The next result gives a criterion for Q to be an injective R -module (a result due to Baer, who introduced the notion of injective modules around 1940), and using it we can give a characterization of injective modules when $R = \mathbb{Z}$ (or, more generally, when R is a P.I.D.). Recall that a $\mathbb{Z}$ -module A (i.e., an abelian group, written additively) is said to be *divisible* if $A = nA$ for all nonzero integers n . For example, both $\mathbb{Q}$ and $\mathbb{Q}/\mathbb{Z}$ are divisible (cf. Exercises 18 and 19 in Section 2.4 and Exercise 15 in Section 3.1).

Proposition 36. Let Q be an R -module.

- (1) (*Baer's Criterion*) The module Q is injective if and only if for every left ideal I of R any R -module homomorphism $g : I \rightarrow Q$ can be extended to an R -module homomorphism $G : R \rightarrow Q$.
- (2) If R is a P.I.D. then Q is injective if and only if $rQ = Q$ for every nonzero $r \in R$. In particular, a $\mathbb{Z}$ -module is injective if and only if it is divisible. When R is a P.I.D., quotient modules of injective R -modules are again injective.

Proof: If Q is injective and $g : I \rightarrow Q$ is an R -module homomorphism from the nonzero ideal I of R into Q , then g can be extended to an R -module homomorphism from R into Q by Proposition 34(2) applied to the exact sequence $0 \rightarrow I \rightarrow R$, which proves the “only if” portion of (1). Suppose conversely that every homomorphism $g : I \rightarrow Q$ can be lifted to a homomorphism $G : R \rightarrow Q$. To show that Q is injective we must show that if $0 \rightarrow L \rightarrow M$ is exact and $f : L \rightarrow Q$ is an R -module homomorphism then there is a lift $F : M \rightarrow Q$ extending f . If $\mathcal{S}$ is the collection (f', L') of lifts $f' : L' \rightarrow Q$ of f to a submodule L' of M containing L , then the ordering $(f', L') \leq (f'', L'')$ if $L' \subseteq L''$ and $f'' = f'$ on L' partially orders $\mathcal{S}$. Since $\mathcal{S} \neq \emptyset$, by Zorn's Lemma there is a maximal element (F, M') in $\mathcal{S}$. The map $F : M' \rightarrow Q$ is a lift of f and it suffices to show that $M' = M$. Suppose that there is some element $m \in M$ not contained in M' and let $I = \{r \in R \mid rm \in M'\}$. It is easy to check that I is a left ideal in R , and the map $g : I \rightarrow Q$ defined by $g(x) = F(xm)$ is an R -module homomorphism from I to Q . By hypothesis, there is a lift $G : R \rightarrow Q$ of g . Consider the submodule $M' + Rm$ of M , and define the map $F' : M' + Rm \rightarrow Q$ by $F'(m' + rm) = F(m') + G(r)$. If $m_1 + r_1m = m_2 + r_2m$ then $(r_1 - r_2)m = m_2 - m_1$

shows that $r_1 - r_2 \in I$, so that

$$G(r_1 - r_2) = g(r_1 - r_2) = F((r_1 - r_2)m) = F(m_2 - m_1),$$

and so $F(m_1) + G(r_1) = F(m_2) + G(r_2)$. Hence F' is well defined and it is then immediate that F' is an R -module homomorphism extending f to $M' + Rm$. This contradicts the maximality of M' , so that $M' = M$, which completes the proof of (1).

To prove (2), suppose R is a P.I.D. Any nonzero ideal I of R is of the form $I = (r)$ for some nonzero element r of R . An R -module homomorphism $f : I \rightarrow Q$ is completely determined by the image $f(r) = q$ in Q . This homomorphism can be extended to a homomorphism $F : R \rightarrow Q$ if and only if there is an element q' in Q with $F(1) = q'$ satisfying $q = f(r) = F(r) = rq'$. It follows that Baer's criterion for Q is satisfied if and only if $rQ = Q$, which proves the first two statements in (2). The final statement follows since a quotient of a module Q with $rQ = Q$ for all $r \neq 0$ in R has the same property.

Examples

- (1) Since $\mathbb{Z}$ is not divisible, $\mathbb{Z}$ is not an injective $\mathbb{Z}$ -module. This also follows from the fact that the exact sequence $0 \rightarrow \mathbb{Z} \xrightarrow{2} \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow 0$ corresponding to multiplication by 2 does not split.
- (2) The rational numbers $\mathbb{Q}$ is an injective $\mathbb{Z}$ -module.
- (3) The quotient $\mathbb{Q}/\mathbb{Z}$ of the injective $\mathbb{Z}$ -module $\mathbb{Q}$ is an injective $\mathbb{Z}$ -module.
- (4) It is immediate that a direct sum of divisible $\mathbb{Z}$ -modules is again divisible, hence a direct sum of injective $\mathbb{Z}$ -modules is again injective. For example, $\mathbb{Q} \oplus \mathbb{Q}/\mathbb{Z}$ is an injective $\mathbb{Z}$ -module. (See also Exercise 4).
- (5) We shall see in Chapter 12 that no nonzero finitely generated $\mathbb{Z}$ -module is injective.
- (6) Suppose that the ring R is an integral domain. An R -module A is said to be a *divisible* R -module if $rA = A$ for every nonzero $r \in R$. The proof of Proposition 36 shows that in this case an injective R -module is divisible.
- (7) We shall see in Section 11.1 that if $R = F$ is a field then every F -module is injective.
- (8) We shall see in Part VI that if F is any field and $n \in \mathbb{Z}^+$ then the ring $R = M_n(F)$ of all $n \times n$ matrices with entries from F has the property that every R -module is injective (and also projective). We shall also see that if G is a finite group of order n and $n \neq 0$ in the field F then the group ring FG also has the property that every module is injective (and also projective).

Corollary 37. Every $\mathbb{Z}$ -module is a submodule of an injective $\mathbb{Z}$ -module.

Proof: Let M be a $\mathbb{Z}$ -module and let A be any set of $\mathbb{Z}$ -module generators of M . Let $\mathcal{F} = F(A)$ be the free $\mathbb{Z}$ -module on the set A . Then by Theorem 6 there is a surjective $\mathbb{Z}$ -module homomorphism from $\mathcal{F}$ to M and if $\mathcal{K}$ denotes the kernel of this homomorphism then $\mathcal{K}$ is a $\mathbb{Z}$ -submodule of $\mathcal{F}$ and we can identify $M = \mathcal{F}/\mathcal{K}$. Let $\mathcal{Q}$ be the free $\mathbb{Q}$ -module on the set A . Then $\mathcal{Q}$ is a direct sum of a number of copies of $\mathbb{Q}$, so is a divisible, hence (by Proposition 36) injective, $\mathbb{Z}$ -module containing $\mathcal{F}$. Then $\mathcal{K}$ is also a $\mathbb{Z}$ -submodule of $\mathcal{Q}$, so the quotient $\mathcal{Q}/\mathcal{K}$ is injective, again by Proposition 36. Since $M = \mathcal{F}/\mathcal{K} \subseteq \mathcal{Q}/\mathcal{K}$, it follows that M is contained in an injective $\mathbb{Z}$ -module.

Corollary 37 can be used to prove the following more general version valid for arbitrary R -modules. This theorem is the injective analogue of the results in Theorem 6 and Corollary 31 showing that every R -module is a quotient of a projective R -module.

Theorem 38. Let R be a ring with 1 and let M be an R -module. Then M is contained in an injective R -module.

Proof: A proof is outlined in Exercises 15 to 17.

It is possible to prove a sharper result than Theorem 38, namely that there is a *minimal* injective R -module H containing M in the sense that any injective map of M into an injective R -module Q factors through H . More precisely, if $M \subseteq Q$ for an injective R -module Q then there is an injection $\iota : H \hookrightarrow Q$ that restricts to the identity map on M ; using ι to identify H as a subset of Q we have $M \subseteq H \subseteq Q$. (cf. Theorem 57.13 in *Representation Theory of Finite Groups and Associative Algebras* by C. Curtis and I. Reiner, John Wiley & Sons, 1966). This module H is called the *injective hull* or *injective envelope* of M . The universal property of the injective hull of M with respect to inclusions of M into injective R -modules should be compared to the universal property with respect to homomorphisms of M of the free module $F(A)$ on a set of generators A for M in Theorem 6. For example, the injective hull of $\mathbb{Z}$ is $\mathbb{Q}$, and the injective hull of any field is itself (cf. the exercises).

Flat Modules and $D \otimes_R _$

We now consider the behavior of extensions $0 \longrightarrow L \xrightarrow{\psi} M \xrightarrow{\varphi} N \longrightarrow 0$ of R -modules with respect to tensor products.

Suppose that D is a *right* R -module. For any homomorphism $f : X \rightarrow Y$ of left R -modules we obtain a homomorphism $1 \otimes f : D \otimes_R X \rightarrow D \otimes_R Y$ of abelian groups (Theorem 13). If in addition D is an (S, R) -bimodule (for example, when $S = R$ is commutative and D is given the standard (R, R) -bimodule structure as in Section 4), then $1 \otimes f$ is a homomorphism of left S -modules. Put another way,

$$D \otimes_R _ : X \longrightarrow D \otimes_R X$$

is a *covariant functor* from the category of left R -modules to the category of abelian groups (respectively, to the category of left S -modules when D is an (S, R) -bimodule), cf. Appendix II. In a similar way, if D is a left R -module then $_ \otimes_R D$ is a covariant functor from the category of right R -modules to the category of abelian groups (respectively, to the category of right S -modules when D is an (R, S) -bimodule). Note that, unlike Hom , the tensor product is covariant in both variables, and we shall therefore concentrate on $D \otimes_R _$, leaving as an exercise the minor alterations necessary for $_ \otimes_R D$.

We have already seen examples where the map $1 \otimes \psi : D \otimes_R L \rightarrow D \otimes_R M$ induced by an injective map $\psi : L \hookrightarrow M$ is no longer injective (for example the injection $\mathbb{Z} \hookrightarrow \mathbb{Q}$ of $\mathbb{Z}$ -modules induces the zero map from $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z} = \mathbb{Z}/2\mathbb{Z}$ to $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Q} = 0$). On the other hand, suppose that $\varphi : M \rightarrow N$ is a surjective R -module homomorphism. The tensor product $D \otimes_R N$ is generated as an abelian group by the simple tensors $d \otimes n$ for $d \in D$ and $n \in N$. The surjectivity of φ implies that $n = \varphi(m)$ for some $m \in M$, and then $1 \otimes \varphi(d \otimes m) = d \otimes \varphi(m) = d \otimes n$ shows that $1 \otimes \varphi$ is a surjective homomorphism of abelian groups from $D \otimes_R M$ to $D \otimes_R N$. This proves most of the following theorem.

Theorem 39. Suppose that D is a right R -module and that L , M and N are left R -modules. If

$$0 \longrightarrow L \xrightarrow{\psi} M \xrightarrow{\varphi} N \longrightarrow 0 \quad \text{is exact,}$$

then the associated sequence of abelian groups

$$D \otimes_R L \xrightarrow{1 \otimes \psi} D \otimes_R M \xrightarrow{1 \otimes \varphi} D \otimes_R N \longrightarrow 0 \quad \text{is exact.} \quad (10.13)$$

If D is an (S, R) -bimodule then (13) is an exact sequence of left S -modules. In particular, if $S = R$ is a commutative ring, then (13) is an exact sequence of R -modules with respect to the standard R -module structures. The map $1 \otimes \varphi$ is not in general injective, i.e., the sequence (13) cannot in general be extended to a short exact sequence.

The sequence (13) is exact for *all* right R -modules D if and only if

$$L \xrightarrow{\psi} M \xrightarrow{\varphi} N \longrightarrow 0 \quad \text{is exact.}$$

Proof: For the first statement it remains to prove the exactness of (13) at $D \otimes_R M$. Since $\varphi \circ \psi = 0$, we have

$$(1 \otimes \varphi) \left(\sum d_i \otimes \psi(l_i) \right) = \sum d_i \otimes (\varphi \circ \psi(l_i)) = 0$$

and it follows that $\text{image}(1 \otimes \psi) \subseteq \ker(1 \otimes \varphi)$. In particular, there is a natural projection $\pi : (D \otimes_R M) / \text{image}(1 \otimes \psi) \rightarrow (D \otimes_R M) / \ker(1 \otimes \varphi) = D \otimes_R N$. The composite of the two projection homomorphisms

$$D \otimes_R M \rightarrow (D \otimes_R M) / \text{image}(1 \otimes \psi) \xrightarrow{\pi} D \otimes_R N$$

is the quotient of $D \otimes_R M$ by $\ker(1 \otimes \varphi)$, so is just the map $1 \otimes \varphi$. We shall show that π is an isomorphism, which will show that the kernel of $1 \otimes \varphi$ is just the kernel of the first projection above, i.e., $\text{image}(1 \otimes \psi)$, giving the exactness of (13) at $D \otimes_R M$. To see that π is an isomorphism we define an inverse map. First define $\pi' : D \times N \rightarrow (D \otimes_R M) / \text{image}(1 \otimes \psi)$ by $\pi'((d, n)) = d \otimes m$ for any $m \in M$ with $\varphi(m) = n$. Note that this is well defined: any other element $m' \in M$ mapping to n differs from m by an element in $\ker \varphi = \text{image } \psi$, i.e., $m' = m + \psi(l)$ for some $l \in L$, and $d \otimes \psi(l) \in \text{image}(1 \otimes \psi)$. It is easy to check that π' is a balanced map, so induces a homomorphism $\tilde{\pi} : D \times N \rightarrow (D \otimes_R M) / \text{image}(1 \otimes \psi)$ with $\tilde{\pi}(d \otimes n) = d \otimes m$. Then $\tilde{\pi} \circ \pi(d \otimes m) = \tilde{\pi}(d \otimes \varphi(m)) = d \otimes m$ shows that $\tilde{\pi} \circ \pi = 1$. Similarly, $\pi \circ \tilde{\pi} = 1$, so that π and $\tilde{\pi}$ are inverse isomorphisms, completing the proof that (13) is exact. Note also that the injectivity of ψ was not required for the proof.

Finally, suppose (13) is exact for every right R -module D . In general, $R \otimes_R X \cong X$ for any left R -module X (Example 1 following Corollary 9). Taking $D = R$ the exactness of the sequence $L \xrightarrow{\psi} M \xrightarrow{\varphi} N \rightarrow 0$ follows.

By Theorem 39, the sequence

$$0 \longrightarrow D \otimes_R L \xrightarrow{1 \otimes \psi} D \otimes_R M \xrightarrow{1 \otimes \varphi} D \otimes_R N \longrightarrow 0$$

is not in general exact since $1 \otimes \psi$ need not be injective. If $0 \rightarrow L \xrightarrow{\psi} M \xrightarrow{\varphi} N \rightarrow 0$ is a *split* short exact sequence, however, then since tensor products commute with direct sums by Theorem 17, it follows that

$$0 \longrightarrow D \otimes_R L \xrightarrow{1 \otimes \psi} D \otimes_R M \xrightarrow{1 \otimes \varphi} D \otimes_R N \longrightarrow 0$$

is also a split short exact sequence.

The following result relating to modules D having the property that (13) can always be extended to a short exact sequence is immediate from Theorem 39:

Proposition 40. Let A be a right R -module. Then the following are equivalent:

- (1) For any left R -modules L , M , and N , if

$$0 \longrightarrow L \xrightarrow{\psi} M \xrightarrow{\varphi} N \longrightarrow 0$$

is a short exact sequence, then

$$0 \longrightarrow A \otimes_R L \xrightarrow{1 \otimes \psi} A \otimes_R M \xrightarrow{1 \otimes \varphi} A \otimes_R N \longrightarrow 0$$

is also a short exact sequence.

- (2) For any left R -modules L and M , if $0 \rightarrow L \xrightarrow{\psi} M$ is an exact sequence of left R -modules (i.e., $\psi : L \rightarrow M$ is injective) then $0 \rightarrow A \otimes_R L \xrightarrow{1 \otimes \psi} A \otimes_R M$ is an exact sequence of abelian groups (i.e., $1 \otimes \psi : A \otimes_R L \rightarrow A \otimes_R M$ is injective).

Definition. A right R -module A is called *flat* if it satisfies either of the two equivalent conditions of Proposition 40.

For a fixed right R -module D , the first part of Theorem 39 is referred to by saying that the functor $D \otimes_R _$ is *right exact*.

Corollary 41. If D is a right R -module, then the functor $D \otimes_R _$ from the category of left R -modules to the category of abelian groups is right exact. If D is an (S, R) -bimodule (for example when $S = R$ is commutative and D is given the standard R -module structure), then $D \otimes_R _$ is a right exact functor from the category of left R -modules to the category of left S -modules. The functor is exact if and only if D is a flat R -module.

We have already seen some flat modules:

Corollary 42. Free modules are flat; more generally, projective modules are flat.

Proof: To show that the free R -module F is flat it suffices to show that for any injective map $\psi : L \rightarrow M$ of R -modules L and M the induced map $1 \otimes \psi : F \otimes_R L \rightarrow F \otimes_R M$ is also injective. Suppose first that $F \cong R^n$ is a finitely generated free R -module. In this case $F \otimes_R L = R^n \otimes_R L \cong L^n$ since $R \otimes_R L \cong L$ and tensor products commute with direct sums. Similarly $F \otimes_R M \cong M^n$ and under these isomorphisms

the map $1 \otimes \psi : F \otimes_R L \rightarrow F \otimes_R M$ is just the natural map of L^n to M^n induced by the inclusion ψ in each component. In particular, $1 \otimes \psi$ is injective and it follows that any finitely generated free module is flat. Suppose now that F is an arbitrary free module and that the element $\sum f_i \otimes l_i \in F \otimes_R L$ is mapped to 0 by $1 \otimes \psi$. This means that the element $\sum (f_i, \psi(l_i))$ can be written as a sum of generators as in equation (6) in the previous section in the free group on $F \times M$. Since this sum of elements is finite, all of the first coordinates of the resulting equation lie in some finitely generated free submodule F' of F . Then this equation implies that $\sum f_i \otimes l_i \in F' \otimes_R L$ is mapped to 0 in $F' \otimes_R M$. Since F' is a finitely generated free module, the injectivity we proved above shows that $\sum f_i \otimes l_i$ is 0 in $F' \otimes_R L$ and so also in $F \otimes_R L$. It follows that $1 \otimes \psi$ is injective and hence that F is flat.

Suppose now that P is a projective module. Then P is a direct summand of a free module F (Proposition 30), say $F = P \oplus P'$. If $\psi : L \rightarrow M$ is injective then $1 \otimes \psi : F \otimes_R L \rightarrow F \otimes_R M$ is also injective by what we have already shown. Since $F = P \oplus P'$ and tensor products commute with direct sums, this shows that

$$1 \otimes \psi : (P \otimes_R L) \oplus (P' \otimes_R L) \rightarrow (P \otimes_R M) \oplus (P' \otimes_R M)$$

is injective. Hence $1 \otimes \psi : P \otimes_R L \rightarrow P \otimes_R M$ is injective, proving that P is flat.

Examples

- (1) Since $\mathbb{Z}$ is a projective $\mathbb{Z}$ -module it is flat. The example before Theorem 39 shows that $\mathbb{Z}/2\mathbb{Z}$ is not a flat $\mathbb{Z}$ -module.
- (2) The $\mathbb{Z}$ -module $\mathbb{Q}$ is a flat $\mathbb{Z}$ -module, as follows. Suppose $\psi : L \rightarrow M$ is an injective map of $\mathbb{Z}$ -modules. Every element of $\mathbb{Q} \otimes_{\mathbb{Z}} L$ can be written in the form $(1/d) \otimes l$ for some nonzero integer d and some $l \in L$ (Exercise 7 in Section 4). If $(1/d) \otimes l$ is in the kernel of $1 \otimes \psi$ then $(1/d) \otimes \psi(l)$ is 0 in $\mathbb{Q} \otimes_{\mathbb{Z}} M$. By Exercise 8 in Section 4 this means $c\psi(l) = 0$ in M for some nonzero integer c . Then $\psi(c \cdot l) = 0$, and the injectivity of ψ implies $c \cdot l = 0$ in L . But this implies that $(1/d) \otimes l = (1/cd) \otimes (c \cdot l) = 0$ in L , which shows that $1 \otimes \psi$ is injective.
- (3) The $\mathbb{Z}$ -module $\mathbb{Q}/\mathbb{Z}$ is injective (by Proposition 36), but is not flat: the injective map $\psi(z) = 2z$ from $\mathbb{Z}$ to $\mathbb{Z}$ does not remain injective after tensoring with $\mathbb{Q}/\mathbb{Z}$ ($1 \otimes \psi : \mathbb{Q}/\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z} \rightarrow \mathbb{Q}/\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}$ has the nonzero element $(\frac{1}{2} + \mathbb{Z}) \otimes 1$ in its kernel — identifying $\mathbb{Q}/\mathbb{Z} = \mathbb{Q}/\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}$ this is the statement that multiplication by 2 has the element $1/2$ in its kernel).
- (4) The direct sum of flat modules is flat (Exercise 5). In particular, $\mathbb{Q} \oplus \mathbb{Z}$ is flat. This module is neither projective nor injective (since $\mathbb{Q}$ is not projective by Exercise 8 and $\mathbb{Z}$ is not injective by Proposition 36 (cf. Exercises 3 and 4)).

We close this section with an important relation between Hom and tensor products:

Theorem 43. (Adjoint Associativity) Let R and S be rings, let A be a right R -module, let B be an (R, S) -bimodule and let C be a right S -module. Then there is an isomorphism of abelian groups:

$$\text{Hom}_S(A \otimes_R B, C) \cong \text{Hom}_R(A, \text{Hom}_S(B, C))$$

(the homomorphism groups are right module homomorphisms—note that $\text{Hom}_S(B, C)$ has the structure of a right R -module, cf. the exercises). If $R = S$ is commutative this is an isomorphism of R -modules with the standard R -module structures.

Proof: Suppose $\varphi : A \otimes_R B \rightarrow C$ is a homomorphism. For any fixed $a \in A$ define the map $\Phi(a)$ from B to C by $\Phi(a)(b) = \varphi(a \otimes b)$. It is easy to check that $\Phi(a)$ is a homomorphism of right S -modules and that the map Φ from A to $\text{Hom}_S(B, C)$ given by mapping a to $\Phi(a)$ is a homomorphism of right R -modules. Then $f(\varphi) = \Phi$ defines a group homomorphism from $\text{Hom}_S(A \otimes_R B, C)$ to $\text{Hom}_R(A, \text{Hom}_S(B, C))$. Conversely, suppose $\Phi : A \rightarrow \text{Hom}_S(B, C)$ is a homomorphism. The map from $A \times B$ to C defined by mapping (a, b) to $\Phi(a)(b)$ is an R -balanced map, so induces a homomorphism φ from $A \otimes_R B$ to C . Then $g(\Phi) = \varphi$ defines a group homomorphism inverse to f and gives the isomorphism in the theorem.

As a first application of Theorem 43 we give an alternate proof of the first result in Theorem 39 that the tensor product is right exact in the case where $S = R$ is a commutative ring. If $0 \rightarrow L \rightarrow M \rightarrow N \rightarrow 0$ is exact, then by Theorem 33 the sequence

$$0 \rightarrow \text{Hom}_R(N, E) \rightarrow \text{Hom}_R(M, E) \rightarrow \text{Hom}_R(L, E)$$

is exact for every R -module E . Then by Theorem 28, the sequence

$$0 \rightarrow \text{Hom}_R(D, \text{Hom}_R(N, E)) \rightarrow \text{Hom}_R(D, \text{Hom}_R(M, E)) \rightarrow \text{Hom}_R(D, \text{Hom}_R(L, E))$$

is exact for all D and all E . By adjoint associativity, this means the sequence

$$0 \rightarrow \text{Hom}_R(D \otimes_R N, E) \rightarrow \text{Hom}_R(D \otimes_R M, E) \rightarrow \text{Hom}_R(D \otimes_R L, E)$$

is exact for any D and all E . Then, by the second part of Theorem 33, it follows that the sequence

$$D \otimes_R L \rightarrow D \otimes_R M \rightarrow D \otimes_R N \rightarrow 0$$

is exact for all D , which is the right exactness of the tensor product.

As a second application of Theorem 43 we prove that the tensor product of two projective modules over a commutative ring R is again projective (see also Exercise 9 for a more direct proof).

Corollary 44. If R is commutative then the tensor product of two projective R -modules is projective.

Proof: Let P_1 and P_2 be projective modules. Then by Corollary 32, $\text{Hom}_R(P_2, _)$ is an exact functor from the category of R -modules to the category of R -modules. Then the composition $\text{Hom}_R(P_1, \text{Hom}_R(P_2, _))$ is an exact functor by the same corollary. By Theorem 43 this means that $\text{Hom}_R(P_1 \otimes_R P_2, _)$ is an exact functor on R -modules. It follows again from Corollary 32 that $P_1 \otimes_R P_2$ is projective.

Summary

Each of the functors $\text{Hom}_R(A, _)$, $\text{Hom}_R(_, A)$, and $A \otimes_R _$, map left R -modules to abelian groups; the functor $_ \otimes_R A$ maps right R -modules to abelian groups. When R is commutative all four functors map R -modules to R -modules.

- (1) Let A be a left R -module. The functor $\text{Hom}_R(A, _)$ is covariant and left exact; the module A is projective if and only if $\text{Hom}_R(A, _)$ is exact (i.e., is also right exact).

- (2) Let A be a left R -module. The functor $\text{Hom}_R(_, A)$ is contravariant and left exact; the module A is injective if and only if $\text{Hom}_R(_, A)$ is exact.
- (3) Let A be a right R -module. The functor $A \otimes_R _$ is covariant and right exact; the module A is flat if and only if $A \otimes_R _$ is exact (i.e., is also left exact).
- (4) Let A be a left R -module. The functor $_ \otimes_R A$ is covariant and right exact; the module A is flat if and only if $_ \otimes_R A$ is exact.
- (5) Projective modules are flat. The $\mathbb{Z}$ -module $\mathbb{Q}/\mathbb{Z}$ is injective but not flat. The $\mathbb{Z}$ -module $\mathbb{Z} \oplus \mathbb{Q}$ is flat but neither projective nor injective.

EXERCISES

Let R be a ring with 1.

1. Suppose that

$$\begin{array}{ccccc} A & \xrightarrow{\psi} & B & \xrightarrow{\varphi} & C \\ \alpha \downarrow & & \beta \downarrow & & \gamma \downarrow \\ A' & \xrightarrow{\psi'} & B' & \xrightarrow{\varphi'} & C' \end{array}$$

is a commutative diagram of groups and that the rows are exact. Prove that

- (a) if φ and α are surjective, and β is injective then γ is injective. [If $c \in \ker \gamma$, show there is a $b \in B$ with $\varphi(b) = c$. Show that $\varphi'(\beta(b)) = 0$ and deduce that $\beta(b) = \psi'(a')$ for some $a' \in A'$. Show there is an $a \in A$ with $\alpha(a) = a'$ and that $\beta(\psi(a)) = \beta(b)$. Conclude that $b = \psi(a)$ and hence $c = \varphi(b) = 0$.]
- (b) if ψ' , α , and γ are injective, then β is injective,
- (c) if φ , α , and γ are surjective, then β is surjective,
- (d) if β is injective, α and γ are surjective, then γ is injective,
- (e) if β is surjective, γ and ψ' are injective, then α is surjective.

2. Suppose that

$$\begin{array}{ccccccc} A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & D \\ \alpha \downarrow & & \beta \downarrow & & \gamma \downarrow & & \delta \downarrow \\ A' & \longrightarrow & B' & \longrightarrow & C' & \longrightarrow & D' \end{array}$$

is a commutative diagram of groups, and that the rows are exact. Prove that

- (a) if α is surjective, and β , δ are injective, then γ is injective.
- (b) if δ is injective, and α , γ are surjective, then β is surjective.
3. Let P_1 and P_2 be R -modules. Prove that $P_1 \oplus P_2$ is a projective R -module if and only if both P_1 and P_2 are projective.
4. Let Q_1 and Q_2 be R -modules. Prove that $Q_1 \oplus Q_2$ is an injective R -module if and only if both Q_1 and Q_2 are injective.
5. Let A_1 and A_2 be R -modules. Prove that $A_1 \oplus A_2$ is a flat R -module if and only if both A_1 and A_2 are flat. More generally, prove that an arbitrary direct sum $\sum A_i$ of R -modules is flat if and only if each A_i is flat. [Use the fact that tensor product commutes with arbitrary direct sums.]
6. Prove that the following are equivalent for a ring R :
- Every R -module is projective.
 - Every R -module is injective.

7. Let A be a nonzero finite abelian group.
 - (a) Prove that A is not a projective $\mathbb{Z}$ -module.
 - (b) Prove that A is not an injective $\mathbb{Z}$ -module.
8. Let Q be a nonzero divisible $\mathbb{Z}$ -module. Prove that Q is not a projective $\mathbb{Z}$ -module. Deduce that the rational numbers $\mathbb{Q}$ is not a projective $\mathbb{Z}$ -module. [Show first that if F is any free module then $\bigcap_{n=1}^{\infty} nF = 0$ (use a basis of F to prove this). Now suppose to the contrary that Q is projective and derive a contradiction from Proposition 30(4).]
9. Assume R is commutative with 1.
 - (a) Prove that the tensor product of two free R -modules is free. [Use the fact that tensor products commute with direct sums.]
 - (b) Use (a) to prove that the tensor product of two projective R -modules is projective.
10. Let R and S be rings with 1 and let M and N be left R -modules. Assume also that M is an (R, S) -bimodule.
 - (a) For $s \in S$ and for $\varphi \in \text{Hom}_R(M, N)$ define $(s\varphi) : M \rightarrow N$ by $(s\varphi)(m) = \varphi(ms)$. Prove that $s\varphi$ is a homomorphism of left R -modules, and that this action of S on $\text{Hom}_R(M, N)$ makes it into a *left* S -module.
 - (b) Let $S = R$ and let $M = R$ (considered as an (R, R) -bimodule by left and right ring multiplication on itself). For each $n \in N$ define $\varphi_n : R \rightarrow N$ by $\varphi_n(r) = rn$, i.e., φ_n is the unique R -module homomorphism mapping 1_R to n . Show that $\varphi_n \in \text{Hom}_R(R, N)$. Use part (a) to show that the map $n \mapsto \varphi_n$ is an isomorphism of left R -modules: $N \cong \text{Hom}_R(R, N)$.
 - (c) Deduce that if N is a free (respectively, projective, injective, flat) left R -module, then $\text{Hom}_R(R, N)$ is also a free (respectively, projective, injective, flat) left R -module.
11. Let R and S be rings with 1 and let M and N be left R -modules. Assume also that N is an (R, S) -bimodule.
 - (a) For $s \in S$ and for $\varphi \in \text{Hom}_R(M, N)$ define $(\varphi s) : M \rightarrow N$ by $(\varphi s)(m) = \varphi(m)s$. Prove that φs is a homomorphism of left R -modules, and that this action of S on $\text{Hom}_R(M, N)$ makes it into a *right* S -module. Deduce that $\text{Hom}_R(M, R)$ is a right R -module, for any R -module M —called the *dual module* to M .
 - (b) Let $N = R$ be considered as an (R, R) -bimodule as usual. Under the action defined in part (a) show that the map $r \mapsto \varphi_r$ is an isomorphism of right R -modules: $\text{Hom}_R(R, R) \cong R$, where φ_r is the homomorphism that maps 1_R to r . Deduce that if M is a finitely generated free left R -module, then $\text{Hom}_R(M, R)$ is a free right R -module of the same rank. (cf. also Exercise 13.)
 - (c) Show that if M is a finitely generated projective R -module then its dual module $\text{Hom}_R(M, R)$ is also projective.
12. Let A be an R -module, let I be any nonempty index set and for each $i \in I$ let B_i be an R -module. Prove the following isomorphisms of abelian groups; when R is commutative prove also that these are R -module isomorphisms. (Arbitrary direct sums and direct products of modules are introduced in Exercise 20 of Section 3.)
 - (a) $\text{Hom}_R(\bigoplus_{i \in I} B_i, A) \cong \prod_{i \in I} \text{Hom}_R(B_i, A)$
 - (b) $\text{Hom}_R(A, \prod_{i \in I} B_i) \cong \prod_{i \in I} \text{Hom}_R(A, B_i)$.
13. (a) Show that the dual of the free $\mathbb{Z}$ -module with countable basis is not free. [Use the preceding exercise and Exercise 24, Section 3.] (See also Exercise 5 in Section 11.3.)
 - (b) Show that the dual of the free $\mathbb{Z}$ -module with countable basis is also not projective. [You may use the fact that any submodule of a free $\mathbb{Z}$ -module is free.]
14. Let $0 \rightarrow L \xrightarrow{\psi} M \xrightarrow{\varphi} N \rightarrow 0$ be a sequence of R -modules.

- (a) Prove that the associated sequence

$$0 \longrightarrow \operatorname{Hom}_R(D, L) \xrightarrow{\psi'} \operatorname{Hom}_R(D, M) \xrightarrow{\varphi'} \operatorname{Hom}_R(D, N) \longrightarrow 0$$

is a short exact sequence of abelian groups for all R -modules D if and only if the original sequence is a split short exact sequence. [To show the sequence splits, take $D = N$ and show the lift of the identity map in $\operatorname{Hom}_R(N, N)$ to $\operatorname{Hom}_R(N, M)$ is a splitting homomorphism for φ .]

- (b) Prove that the associated sequence

$$0 \longrightarrow \operatorname{Hom}_R(N, D) \xrightarrow{\varphi'} \operatorname{Hom}_R(M, D) \xrightarrow{\psi'} \operatorname{Hom}_R(L, D) \longrightarrow 0$$

is a short exact sequence of abelian groups for all R -modules D if and only if the original sequence is a split short exact sequence.

15. Let M be a left R -module where R is a ring with 1.

- (a) Show that $\operatorname{Hom}_{\mathbb{Z}}(R, M)$ is a left R -module under the action $(r\varphi)(r') = \varphi(r'r)$ (see Exercise 10).

- (b) Suppose that $0 \rightarrow A \xrightarrow{\psi} B$ is an exact sequence of R -modules. Prove that if every homomorphism f from A to M lifts to a homomorphism F from B to M with $f = F \circ \psi$, then every homomorphism f' from A to $\operatorname{Hom}_{\mathbb{Z}}(R, M)$ lifts to a homomorphism F' from B to $\operatorname{Hom}_{\mathbb{Z}}(R, M)$ with $f' = F' \circ \psi$. [Given f' , show that $f(a) = f'(a)(1_R)$ defines a homomorphism of A to M . If F is the associated lift of f to B , show that $F'(b)(r) = F(rb)$ defines a homomorphism from B to $\operatorname{Hom}_{\mathbb{Z}}(R, M)$ that lifts f' .]

- (c) Prove that if Q is an injective R -module then $\operatorname{Hom}_{\mathbb{Z}}(R, Q)$ is also an injective R -module.

16. This exercise proves Theorem 38 that every left R -module M is contained in an injective left R -module.

- (a) Show that M is contained in an injective $\mathbb{Z}$ -module Q . [M is a $\mathbb{Z}$ -module—use Corollary 37.]

- (b) Show that $\operatorname{Hom}_R(R, M) \subseteq \operatorname{Hom}_{\mathbb{Z}}(R, M) \subseteq \operatorname{Hom}_{\mathbb{Z}}(R, Q)$.

- (c) Use the R -module isomorphism $M \cong \operatorname{Hom}_R(R, M)$ (Exercise 10) and the previous exercise to conclude that M is contained in an injective module.

17. This exercise completes the proof of Proposition 34. Suppose that Q is an R -module with the property that every short exact sequence $0 \rightarrow Q \rightarrow M_1 \rightarrow N \rightarrow 0$ splits and suppose

that the sequence $0 \rightarrow L \xrightarrow{\psi} M$ is exact. Prove that every R -module homomorphism f from L to Q can be lifted to an R -module homomorphism F from M to Q with $f = F \circ \psi$. [By the previous exercise, Q is contained in an injective R -module. Use the splitting property together with Exercise 4 (noting that Exercise 4 can be proved using (2) in Proposition 34 as the definition of an injective module).]

18. Prove that the injective hull of the $\mathbb{Z}$ -module $\mathbb{Z}$ is $\mathbb{Q}$. [Let H be the injective hull of $\mathbb{Z}$ and argue that $\mathbb{Q}$ contains an isomorphic copy of H . Use the divisibility of H to show $1/n \in H$ for all nonzero integers n , and deduce that $H = \mathbb{Q}$.]

19. If F is a field, prove that the injective hull of F is F .

20. Prove that the polynomial ring $R[x]$ in the indeterminate x over the commutative ring R is a flat R -module.

21. Let R and S be rings with 1 and suppose M is a right R -module, and N is an (R, S) -bimodule. If M is flat over R and N is flat as an S -module prove that $M \otimes_R N$ is flat as a right S -module.

22. Suppose that R is a commutative ring and that M and N are flat R -modules. Prove that $M \otimes_R N$ is a flat R -module. [Use the previous exercise.]
23. Prove that the (right) module $M \otimes_R S$ obtained by changing the base from the ring R to the ring S (by some homomorphism $f : R \rightarrow S$ with $f(1_R) = 1_S$, cf. Example 6 following Corollary 12 in Section 4) of the flat (right) R -module M is a flat S -module.
24. Prove that A is a flat R -module if and only if for any left R -modules L and M where L is *finitely generated*, then $\psi : L \rightarrow M$ injective implies that also $1 \otimes \psi : A \otimes_R L \rightarrow A \otimes_R M$ is injective. [Use the techniques in the proof of Corollary 42.]
25. (*A Flatness Criterion*) Parts (a)-(c) of this exercise prove that A is a flat R -module if and only if for every finitely generated ideal I of R , the map from $A \otimes_R I \rightarrow A \otimes_R R \cong A$ induced by the inclusion $I \subseteq R$ is again injective (or, equivalently, $A \otimes_R I \cong AI \subseteq A$).
- (a) Prove that if A is flat then $A \otimes_R I \rightarrow A \otimes_R R$ is injective.
- (b) If $A \otimes_R I \rightarrow A \otimes_R R$ is injective for every finitely generated ideal I , prove that $A \otimes_R I \rightarrow A \otimes_R R$ is injective for every ideal I . Show that if K is any submodule of a finitely generated free module F then $A \otimes_R K \rightarrow A \otimes_R F$ is injective. Show that the same is true for any free module F . [Cf. the proof of Corollary 42.]
- (c) Under the assumption in (b), suppose L and M are R -modules and $L \xrightarrow{\psi} M$ is injective. Prove that $A \otimes_R L \xrightarrow{1 \otimes \psi} A \otimes_R M$ is injective and conclude that A is flat. [Write M as a quotient of the free module F , giving a short exact sequence

$$0 \longrightarrow K \longrightarrow F \xrightarrow{f} M \longrightarrow 0.$$

Show that if $J = f^{-1}(\psi(L))$ and $\iota : J \rightarrow F$ is the natural injection, then the diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & K & \longrightarrow & J & \longrightarrow & L \longrightarrow 0 \\ & & \downarrow id & & \downarrow \iota & & \downarrow \psi \\ 0 & \longrightarrow & K & \longrightarrow & F & \longrightarrow & M \longrightarrow 0 \end{array}$$

is commutative with exact rows. Show that the induced diagram

$$\begin{array}{ccccccc} A \otimes_R K & \longrightarrow & A \otimes_R J & \longrightarrow & A \otimes_R L & \longrightarrow & 0 \\ & & \downarrow 1 \otimes \iota & & \downarrow 1 \otimes \psi & & \\ A \otimes_R K & \longrightarrow & A \otimes_R F & \longrightarrow & A \otimes_R M & \longrightarrow & 0 \end{array}$$

is commutative with exact rows. Use (b) to show that $1 \otimes \iota$ is injective, then use Exercise 1 to conclude that $1 \otimes \psi$ is injective.]

- (d) (*A Flatness Criterion for quotients*) Suppose $A = F/K$ where F is flat (e.g., if F is free) and K is an R -submodule of F . Prove that A is flat if and only if $FI \cap K = KI$ for every finitely generated ideal I of R . [Use (a) to prove $F \otimes_R I \cong FI$ and observe the image of $K \otimes_R I$ is KI ; tensor the exact sequence $0 \rightarrow K \rightarrow F \rightarrow A \rightarrow 0$ with I to prove that $A \otimes_R I \cong FI/KI$, and apply the flatness criterion.]
26. Suppose R is a P.I.D. This exercise proves that A is a flat R -module if and only if A is torsion free R -module (i.e., if $a \in A$ is nonzero and $r \in R$, then $ra = 0$ implies $r = 0$).
- (a) Suppose that A is flat and for fixed $r \in R$ consider the map $\psi_r : R \rightarrow R$ defined by multiplication by r : $\psi_r(x) = rx$. If r is nonzero show that ψ_r is an injection. Conclude from the flatness of A that the map from A to A defined by mapping a to ra is injective and that A is torsion free.
- (b) Suppose that A is torsion free. If I is a nonzero ideal of R , then $I = rR$ for some nonzero $r \in R$. Show that the map ψ_r in (a) induces an isomorphism $R \cong I$ of

R -modules and that the composite $R \xrightarrow{\psi} I \xrightarrow{\iota} R$ of ψ_r with the inclusion $\iota : I \subseteq R$ is multiplication by r . Prove that the composite $A \otimes_R R \xrightarrow{1 \otimes \psi_r} A \otimes_R I \xrightarrow{1 \otimes \iota} A \otimes_R R$ corresponds to the map $a \mapsto ra$ under the identification $A \otimes_R R = A$ and that this composite is injective since A is torsion free. Show that $1 \otimes \psi_r$ is an isomorphism and deduce that $1 \otimes \iota$ is injective. Use the previous exercise to conclude that A is flat.

27. Let M , A and B be R -modules.

- (a) Suppose $f : A \rightarrow M$ and $g : B \rightarrow M$ are R -module homomorphisms. Prove that $X = \{(a, b) \mid a \in A, b \in B \text{ with } f(a) = g(b)\}$ is an R -submodule of the direct sum $A \oplus B$ (called the *pullback* or *fiber product* of f and g) and that there is a commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{\pi_2} & B \\ \pi_1 \downarrow & & \downarrow g \\ A & \xrightarrow{f} & M \end{array}$$

where π_1 and π_2 are the natural projections onto the first and second components.

- (b) Suppose $f' : M \rightarrow A$ and $g' : M \rightarrow B$ are R -module homomorphisms. Prove that the quotient Y of $A \oplus B$ by $\{(f'(m), -g'(m)) \mid m \in M\}$ is an R -module (called the *pushout* or *fiber sum* of f' and g') and that there is a commutative diagram

$$\begin{array}{ccc} M & \xrightarrow{g'} & B \\ f' \downarrow & & \downarrow \pi'_2 \\ A & \xrightarrow{\pi'_1} & Y \end{array}$$

where π'_1 and π'_2 are the natural maps to the quotient induced by the maps into the first and second components.

28. (a) (*Schanuel's Lemma*) If $0 \rightarrow K \rightarrow P \xrightarrow{\varphi} M \rightarrow 0$ and $0 \rightarrow K' \rightarrow P' \xrightarrow{\varphi'} M \rightarrow 0$ are exact sequences of R -modules where P and P' are projective, prove $P \oplus K' \cong P' \oplus K$ as R -modules. [Show that there is an exact sequence $0 \rightarrow \ker \pi \rightarrow X \xrightarrow{\pi} P \rightarrow 0$ with $\ker \pi \cong K'$, where X is the fiber product of φ and φ' as in the previous exercise. Deduce that $X \cong P \oplus K'$. Show similarly that $X \cong P' \oplus K$.]
- (b) If $0 \rightarrow M \rightarrow Q \xrightarrow{\psi} L \rightarrow 0$ and $0 \rightarrow M \rightarrow Q' \xrightarrow{\psi'} L' \rightarrow 0$ are exact sequences of R -modules where Q and Q' are injective, prove $Q \oplus L' \cong Q' \oplus L$ as R -modules.

The R -modules M and N are said to be *projectively equivalent* if $M \oplus P \cong N \oplus P'$ for some projective modules P, P' . Similarly, M and N are *injectively equivalent* if $M \oplus Q \cong N \oplus Q'$ for some injective modules Q, Q' . The previous exercise shows K and K' are projectively equivalent and L and L' are injectively equivalent.

Vector Spaces

In this chapter we review the basic theory of finite dimensional vector spaces over an arbitrary field F (some infinite dimensional vector space theory is covered in the exercises). Since the proofs are identical to the corresponding arguments for real vector spaces our treatment is very terse. For the most part we include only those results which are used in other parts of the text so basic topics such as Gauss–Jordan elimination, row echelon forms, methods for finding bases of subspaces, elementary properties of matrices, etc., are not covered or are discussed in the exercises. The reader should therefore consider this chapter as a refresher in linear algebra and as a prelude to field theory and Galois theory. Characteristic polynomials and eigenvalues will be reviewed and treated in a larger context in the next chapter.

11.1 DEFINITIONS AND BASIC THEORY

The terminology for vector spaces is slightly different from that of modules, that is, when the ring R is a field there are different names for many of the properties of R -modules which we defined in the last chapter. The following is a dictionary of these new terms (many of which may already be familiar). The definition of each corresponding vector space property is the same (verbatim) as the module-theoretic definition with the only added assumption being that the ring R is a field (so these definitions are not repeated here).

Terminology for R any Ring

M is an R -module
 m is an element of M
 α is a ring element
 N is a submodule of M
 M/N is a quotient module
 M is a free module of rank n
 M is a finitely generated module
 M is a nonzero cyclic module
 $\varphi : M \rightarrow N$ is an R -module homomorphism
 M and N are isomorphic as R -modules
 the subset A of M generates M
 $M = RA$

Terminology for R a Field

M is a vector space over R
 m is a vector in M
 α is a scalar
 N is a subspace of M
 M/N is a quotient space
 M is a vector space of dimension n
 M is a finite dimensional vector space
 M is a 1-dimensional vector space
 $\varphi : M \rightarrow N$ is a linear transformation
 M and N are isomorphic vector spaces
 the subset A of M spans M
 each element of M is a linear combination
 of elements of A i.e., $M = \text{Span}(A)$

For the remainder of this chapter F is a field and V is a vector space over F .

One of the first results we shall prove about vector spaces is that they are free F -modules, that is, they have bases. Although our arguments treat only the case of finite dimensional spaces, the corresponding result for arbitrary vector spaces is proved in the exercises as an application of Zorn's Lemma. The reader may first wish to review the section in the previous chapter on free modules, especially their properties pertaining to homomorphisms.

Definition.

- (1) A subset S of V is called a set of *linearly independent* vectors if an equation $\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n = 0$ with $\alpha_1, \alpha_2, \dots, \alpha_n \in F$ and $v_1, v_2, \dots, v_n \in S$ implies $\alpha_1 = \alpha_2 = \cdots = \alpha_n = 0$.
- (2) A *basis* of a vector space V is an ordered set of linearly independent vectors which span V . In particular two bases will be considered different even if one is simply a rearrangement of the other. This is sometimes referred to as an *ordered basis*.

Examples

- (1) The space $V = F[x]$ of polynomials in the variable x with coefficients from the field F is in particular a vector space over F . The elements $1, x, x^2, \dots$ are linearly independent by definition (i.e., a polynomial is 0 if and only if all its coefficients are 0). Since these elements also span V by definition, they are a basis for V .
- (2) The collection of solutions of a linear, homogeneous, constant coefficient differential equation (for example, $y'' - 3y' + 2y = 0$) over $\mathbb{C}$ form a vector space over $\mathbb{C}$ since differentiation is a linear operator. Elements of this vector space are linearly independent if they are linearly independent as functions. For example, e^t and e^{2t} are easily seen to be solutions of the equation $y'' - 3y' + 2y = 0$ (differentiation with respect to t). They are linearly independent functions since $ae^t + be^{2t} = 0$ implies $a + b = 0$ (let $t = 0$) and $ae + be^2 = 0$ (let $t = 1$) and the only solution to these two equations is $a = b = 0$. It is a theorem in differential equations that these elements span the set of solutions of this equation, hence are a basis for this space.

Proposition 1. Assume the set $\mathcal{A} = \{v_1, v_2, \dots, v_n\}$ spans the vector space V but no proper subset of $\mathcal{A}$ spans V . Then $\mathcal{A}$ is a basis of V . In particular, any finitely generated (i.e., finitely spanned) vector space over F is a free F -module.

Proof: It is only necessary to prove that $v_1, v_2, \dots, v_n$ are linearly independent. Suppose $\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n = 0$ where not all of the α_i are 0. By reordering, we may assume that $\alpha_1 \neq 0$ and then

$$v_1 = -\frac{1}{\alpha_1}(\alpha_2 v_2 + \cdots + \alpha_n v_n).$$

It follows that $\{v_2, v_3, \dots, v_n\}$ also spans V since any linear combination of $v_1, v_2, \dots, v_n$ can be written as a linear combination of $v_2, v_3, \dots, v_n$ using the equation above. This is a contradiction.

Example

Let F be a field and consider $F[x]/(f(x))$ where $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0$. The ideal $(f(x))$ is a subspace of the vector space $F[x]$ and the quotient $F[x]/(f(x))$ is also a vector space over F . By the Euclidean Algorithm, every polynomial $a(x) \in F[x]$ can be written uniquely in the form $a(x) = q(x)f(x) + r(x)$ where $r(x) \in F[x]$ and $0 \leq \deg r(x) \leq n-1$. Since $q(x)f(x) \in (f(x))$, it follows that every element of the quotient is represented by a polynomial $r(x)$ of degree $\leq n-1$. Two distinct such polynomials cannot be the same in the quotient since this would say their difference (which is a nonzero polynomial of degree at most $n-1$) would be divisible by $f(x)$ (which is of degree n). It follows that the elements $\bar{1}, \bar{x}, \bar{x}^2, \dots, \bar{x}^{n-1}$ (the bar denotes the image of these elements in the quotient, as usual) span $F[x]/(f(x))$ as a vector space over F and that no proper subset of these elements also spans, hence these elements give a basis for $F[x]/(f(x))$.

Corollary 2. Assume the finite set $\mathcal{A}$ spans the vector space V . Then $\mathcal{A}$ contains a basis of V .

Proof: Any subset $\mathcal{B}$ of $\mathcal{A}$ spanning V such that no proper subset of $\mathcal{B}$ also spans V (there clearly exist such subsets) is a basis for V by Proposition 1.

Theorem 3. (A Replacement Theorem) Assume $\mathcal{A} = \{a_1, a_2, \dots, a_n\}$ is a basis for V containing n elements and $\{b_1, b_2, \dots, b_m\}$ is a set of linearly independent vectors in V . Then there is an ordering $a_1, a_2, \dots, a_n$ such that for each $k \in \{1, 2, \dots, m\}$ the set $\{b_1, b_2, \dots, b_k, a_{k+1}, a_{k+2}, \dots, a_n\}$ is a basis of V . In other words, the elements $b_1, b_2, \dots, b_m$ can be used to successively replace the elements of the basis $\mathcal{A}$, still retaining a basis. In particular, $n \geq m$.

Proof: Proceed by induction on k . If $k = 0$ there is nothing to prove, since $\mathcal{A}$ is given as a basis for V . Suppose now that $\{b_1, b_2, \dots, b_k, a_{k+1}, a_{k+2}, \dots, a_n\}$ is a basis for V . Then in particular this is a spanning set, so b_{k+1} is a linear combination:

$$b_{k+1} = \beta_1 b_1 + \cdots + \beta_k b_k + \alpha_{k+1} a_{k+1} + \cdots + \alpha_n a_n. \quad (11.1)$$

Not all of the α_i can be 0, since this would imply b_{k+1} is a linear combination of $b_1, b_2, \dots, b_k$, contrary to the linear independence of these elements. By reordering if necessary, we may assume $\alpha_{k+1} \neq 0$. Then solving this last equation for a_{k+1} as a linear combination of b_{k+1} and $b_1, b_2, \dots, b_k, a_{k+2}, \dots, a_n$ shows

$$\text{Span}\{b_1, b_2, \dots, b_k, b_{k+1}, a_{k+2}, \dots, a_n\} = \text{Span}\{b_1, b_2, \dots, b_k, a_{k+1}, a_{k+2}, \dots, a_n\}$$

and so this is a spanning set for V . It remains to show $b_1, \dots, b_k, b_{k+1}, a_{k+2}, \dots, a_n$ are linearly independent. If

$$\beta_1 b_1 + \cdots + \beta_k b_k + \beta_{k+1} b_{k+1} + \alpha_{k+2} a_{k+2} + \cdots + \alpha_n a_n = 0 \quad (11.2)$$

then substituting for b_{k+1} from the expression for b_{k+1} in equation (1), we obtain a linear combination of $\{b_1, b_2, \dots, b_k, a_{k+1}, a_{k+2}, \dots, a_n\}$ equal to 0, where the coefficient of a_{k+1} is β_{k+1} . Since this last set is a basis by induction, all the coefficients in this linear combination, in particular β_{k+1} , must be 0. But then equation (2) is

$$\beta_1 b_1 + \cdots + \beta_k b_k + \alpha_{k+2} a_{k+2} + \cdots + \alpha_n a_n = 0.$$

Again by the induction hypothesis all the other coefficients must be 0 as well. Thus $\{b_1, b_2, \dots, b_k, b_{k+1}, a_{k+2}, \dots, a_n\}$ is a basis for V , and the induction is complete.

Corollary 4.

- (1) Suppose V has a finite basis with n elements. Any set of linearly independent vectors has $\leq n$ elements. Any spanning set has $\geq n$ elements.
- (2) If V has some finite basis then any two bases of V have the same cardinality.

Proof: (1) This is a restatement of the last result of Theorem 3 and Corollary 2.
 (2) This is immediate from (1) since a basis is both a spanning set and a linearly independent set.

Definition. If V is a finitely generated F -module (i.e., has a finite basis) the cardinality of any basis is called the *dimension* of V and is denoted by $\dim_F V$, or just $\dim V$ when F is clear from the context, and V is said to be *finite dimensional* over F . If V is not finitely generated, V is said to be *infinite dimensional* (written $\dim V = \infty$).

Examples

- (1) The dimension of the space of solutions to the differential equation $y'' - 3y' + 2y = 0$ over $\mathbb{C}$ is 2 (with basis e^t, e^{2t} , for example). In general, it is a theorem in differential equations that the space of solutions of an n^{th} order linear, homogeneous, constant coefficient differential equation of degree n over $\mathbb{C}$ form a vector space over $\mathbb{C}$ of dimension n .
- (2) The dimension over F of the quotient $F[x]/(f(x))$ by the nonzero polynomial $f(x)$ considered above is $n = \deg f(x)$. The space $F[x]$ and its subspace $(f(x))$ are infinite dimensional vector spaces over F .

Corollary 5. (Building-Up Lemma) If A is a set of linearly independent vectors in the finite dimensional space V then there exists a basis of V containing A .

Proof: This is also immediate from Theorem 3, since we can use the elements of A to successively replace the elements of any given basis for V (which exists by the assumption that V is finite dimensional).

Theorem 6. If V is an n dimensional vector space over F , then $V \cong F^n$. In particular, any two finite dimensional vector spaces over F of the same dimension are isomorphic.

Proof: Let $v_1, v_2, \dots, v_n$ be a basis for V . Define the map

$$\varphi : F^n \rightarrow V \quad \text{by} \quad \varphi(\alpha_1, \alpha_2, \dots, \alpha_n) = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n.$$

The map φ is clearly F -linear, is surjective since the v_i span V , and is injective since the v_i are linearly independent, hence is an isomorphism.

Examples

- (1) Let $\mathbb{F}$ be a finite field with q elements and let W be a k -dimensional vector space over $\mathbb{F}$. We show that the number of distinct bases of W is

$$(q^k - 1)(q^k - q)(q^k - q^2) \dots (q^k - q^{k-1}).$$

Every basis of W can be built up as follows. Any nonzero vector w_1 can be the first element of a basis. Since W is isomorphic to $\mathbb{F}^k$, $|W| = q^k$, so there are $q^k - 1$ choices for w_1 . Any vector not in the 1-dimensional space spanned by w_1 is linearly independent from w_1 and so may be chosen for the second basis element, w_2 . A 1-dimensional space is isomorphic to $\mathbb{F}$ and so has q elements. Thus there are $q^k - q$ choices for w_2 . Proceeding in this way one sees that at the i^{th} stage any vector not in the $(i-1)$ -dimensional space spanned by $w_1, w_2, \dots, w_{i-1}$ will be linearly independent from $w_1, w_2, \dots, w_{i-1}$ and so may be chosen for the i^{th} basis vector w_i . An $(i-1)$ -dimensional space is isomorphic to $\mathbb{F}^{i-1}$ and so has q^{i-1} elements. Thus there are $q^k - q^{i-1}$ choices for w_i . The process terminates when w_k is chosen, for then we have k linear independent vectors in a k -dimensional space, hence a basis.

- (2) Let $\mathbb{F}$ be a finite field with q elements and let V be an n -dimensional vector space over $\mathbb{F}$. For each $k \in \{1, 2, \dots, n\}$ we show that the number of subspaces of V of dimension k is

$$\frac{(q^n - 1)(q^n - q) \dots (q^n - q^{k-1})}{(q^k - 1)(q^k - q) \dots (q^k - q^{k-1})}.$$

Any k -dimensional space is spanned by k independent vectors. By arguing as in the preceding example the numerator of the above expression is the number of ways of picking k independent vectors from an n -dimensional space. Two sets of k independent vectors span the same space W if and only if they are both bases of the k -dimensional space W . In order to obtain the formula for the number of distinct subspaces of dimension k we must divide by the number of repetitions, i.e., the number of bases of a fixed k -dimensional space. This factor which appears in the denominator is precisely the number computed in Example 1.

Next, we prove an important relation between the dimension of a subspace, the dimension of its associated quotient space and the dimension of the whole space:

Theorem 7. Let V be a vector space over F and let W be a subspace of V . Then V/W is a vector space with $\dim V = \dim W + \dim V/W$ (where if one side is infinite then both are).

Proof: Suppose W has dimension m and V has dimension n over F and let $w_1, w_2, \dots, w_m$ be a basis for W . By Corollary 5, these linearly independent elements of V can be extended to a basis $w_1, w_2, \dots, w_m, v_{m+1}, \dots, v_n$ of V . The natural surjective projection map of V into V/W maps each w_i to 0. No linear combination of the v_i is mapped to 0, since this would imply this linear combination is an element of W , contrary to the choice of the v_i . Hence, the image V/W of this projection map is isomorphic to the subspace of V spanned by the v_i , hence $\dim V/W = n - m$, which is the theorem when the dimensions are finite. If either side is infinite it is an easy exercise to produce an infinite number of linearly independent vectors showing the other side is also infinite.

Corollary 8. Let $\varphi : V \rightarrow U$ be a linear transformation of vector spaces over F . Then $\ker \varphi$ is a subspace of V , $\varphi(V)$ is a subspace of U and $\dim V = \dim \ker \varphi + \dim \varphi(V)$.

Proof: This follows immediately from Theorem 7. Note that the proof of Theorem 7 is in fact the special case of Corollary 8 where U is the quotient V/W and φ is the natural projection homomorphism.

Corollary 9. Let $\varphi : V \rightarrow W$ be a linear transformation of vector spaces of the same finite dimension. Then the following are equivalent:

- (1) φ is an isomorphism
- (2) φ is injective, i.e., $\ker \varphi = 0$
- (3) φ is surjective, i.e., $\varphi(V) = W$
- (4) φ sends a basis of V to a basis of W .

Proof: The equivalence of these conditions follows from Corollary 8 by counting dimensions.

Definition. If $\varphi : V \rightarrow U$ is a linear transformation of vector spaces over F , $\ker \varphi$ is sometimes called the *null space* of φ and the dimension of $\ker \varphi$ is called the *nullity* of φ . The dimension of $\varphi(V)$ is called the *rank* of φ . If $\ker \varphi = 0$, the transformation is said to be *nonsingular*.

Example

Let F be a finite field with q elements and let V be an n -dimensional vector space over F . Recall that the *general linear group* $GL(V)$ is the group of all nonsingular linear transformations from V to V (the group operation being composition). We show that the order of this group is

$$|GL(V)| = (q^n - 1)(q^n - q)(q^n - q^2) \cdots (q^n - q^{n-1}).$$

To see this, fix a basis $v_1, \dots, v_n$ of V . A linear transformation is nonsingular if and only if it sends this basis to another basis of V . Moreover, if $w_1, \dots, w_n$ is any basis of V , by Theorem 6 in Section 10.3 there is a unique linear transformation which sends v_i to w_i , $1 \leq i \leq n$. Thus the number of nonsingular linear transformations from V to itself equals the number of distinct bases of V . This number, which was computed in Example 1 above (with $k = n$), is the order of $GL(V)$.

EXERCISES

1. Let $V = \mathbb{R}^n$ and let $(a_1, a_2, \dots, a_n)$ be a fixed vector in V . Prove that the collection of elements $(x_1, x_2, \dots, x_n)$ of V with $a_1x_1 + a_2x_2 + \dots + a_nx_n = 0$ is a subspace of V . Determine the dimension of this subspace and find a basis.
2. Let V be the collection of polynomials with coefficients in $\mathbb{Q}$ in the variable x of degree at most 5. Prove that V is a vector space over $\mathbb{Q}$ of dimension 6, with $1, x, x^2, \dots, x^5$ as basis. Prove that $1, 1+x, 1+x+x^2, \dots, 1+x+x^2+x^3+x^4+x^5$ is also a basis for V .

3. Let φ be the linear transformation $\varphi : \mathbb{R}^4 \rightarrow \mathbb{R}^1$ such that

$$\begin{aligned}\varphi((1, 0, 0, 0)) &= 1 & \varphi((1, -1, 0, 0)) &= 0 \\ \varphi((1, -1, 1, 0)) &= 1 & \varphi((1, -1, 1, -1)) &= 0.\end{aligned}$$

Determine $\varphi((a, b, c, d))$.

4. Prove that the space of real-valued functions on the closed interval $[a, b]$ is an infinite dimensional vector space over $\mathbb{R}$, where $a < b$.
5. Prove that the space of continuous real-valued functions on the closed interval $[a, b]$ is an infinite dimensional vector space over $\mathbb{R}$, where $a < b$.
6. Let V be a vector space of finite dimension. If φ is any linear transformation from V to V prove there is an integer m such that the intersection of the image of φ^m and the kernel of φ^m is $\{0\}$.
7. Let φ be a linear transformation from a vector space V of dimension n to itself that satisfies $\varphi^2 = 0$. Prove that the image of φ is contained in the kernel of φ and hence that the rank of φ is at most $n/2$.
8. Let V be a vector space over F and let φ be a linear transformation of the vector space V to itself. A nonzero element $v \in V$ satisfying $\varphi(v) = \lambda v$ for some $\lambda \in F$ is called an *eigenvector* of φ with *eigenvalue* λ . Prove that for any fixed $\lambda \in F$ the collection of eigenvectors of φ with eigenvalue λ together with 0 forms a subspace of V .
9. Let V be a vector space over F and let φ be a linear transformation of the vector space V to itself. Suppose for $i = 1, 2, \dots, k$ that $v_i \in V$ is an eigenvector for φ with eigenvalue $\lambda_i \in F$ (cf. the preceding exercise) and that all the eigenvalues λ_i are distinct. Prove that $v_1, v_2, \dots, v_k$ are linearly independent. [Use induction on k : write a linear dependence relation among the v_i and apply φ to get another linear dependence relation among the v_i involving the eigenvalues — now subtract a suitable multiple of the first linear relation to get a linear dependence relation on fewer elements.] Conclude that any linear transformation on an n -dimensional vector space has at most n distinct eigenvalues.

In the following exercises let V be a vector space of arbitrary dimension over a field F .

10. Prove that any vector space V has a basis (by convention the null set is the basis for the zero space). [Let $\mathcal{S}$ be the set of subsets of V consisting of linearly independent vectors, partially ordered under inclusion; apply Zorn's Lemma to $\mathcal{S}$ and show a maximal element of $\mathcal{S}$ is a basis.]
11. Refine your argument in the preceding exercise to prove that any set of linearly independent vectors of V is contained in a basis of V .
12. If F is a field with a finite or countable number of elements and V is an infinite dimensional vector space over F with basis $\mathcal{B}$, prove that the cardinality of V equals the cardinality of $\mathcal{B}$. Deduce in this case that any two bases of V have the same cardinality.
13. Prove that as vector spaces over $\mathbb{Q}$, $\mathbb{R}^n \cong \mathbb{R}$, for all $n \in \mathbb{Z}^+$ (note that, in particular, this means $\mathbb{R}^n$ and $\mathbb{R}$ are isomorphic as additive abelian groups).
14. Let $\mathcal{A}$ be a basis for the infinite dimensional space V . Prove that V is isomorphic to the *direct sum* of copies of the field F indexed by the set $\mathcal{A}$. Prove that the *direct product* of copies of F indexed by $\mathcal{A}$ is a vector space over F and it has strictly larger dimension than the dimension of V (see the exercises in Section 10.3 for the definitions of direct sum and direct product of infinitely many modules).

11.2 THE MATRIX OF A LINEAR TRANSFORMATION

Throughout this section let V, W be vector spaces over the same field F , let $\mathcal{B} = \{v_1, v_2, \dots, v_n\}$ be an (ordered) basis of V , let $\mathcal{E} = \{w_1, w_2, \dots, w_m\}$ be an (ordered) basis of W and let $\varphi \in \text{Hom}(V, W)$ be a linear transformation from V to W . For each $j \in \{1, 2, \dots, n\}$ write the image of v_j under φ in terms of the basis $\mathcal{E}$:

$$\varphi(v_j) = \sum_{i=1}^m \alpha_{ij} w_i. \quad (11.3)$$

Let $M_{\mathcal{B}}^{\mathcal{E}}(\varphi) = (a_{ij})$ be the $m \times n$ matrix whose i, j entry is α_{ij} (that is, use the coefficients of the w_i 's in the above computation of $\varphi(v_j)$ for the j^{th} column of this matrix). The matrix $M_{\mathcal{B}}^{\mathcal{E}}(\varphi)$ is called the *matrix of φ with respect to the bases $\mathcal{B}, \mathcal{E}$* . The domain basis is the lower and the codomain basis the upper letters appearing after the " M ." Given this matrix, we can recover the linear transformation φ as follows: to compute $\varphi(v)$ for $v \in V$, write v in terms of the basis $\mathcal{B}$:

$$v = \sum_{i=1}^n \alpha_i v_i, \quad \alpha_i \in F,$$

and then calculate the product of the $m \times n$ and $n \times 1$ matrices

$$M_{\mathcal{B}}^{\mathcal{E}}(\varphi) \times \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix} = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_m \end{pmatrix}.$$

The image of v under φ is given by

$$\varphi(v) = \sum_{i=1}^m \beta_i w_i,$$

i.e., the column vector of coordinates of $\varphi(v)$ with respect to the basis $\mathcal{E}$ are obtained by multiplying the matrix $M_{\mathcal{B}}^{\mathcal{E}}(\varphi)$ by the column vector of coordinates of v with respect to the basis $\mathcal{B}$ (sometimes denoted $[\varphi(v)]_{\mathcal{E}} = M_{\mathcal{B}}^{\mathcal{E}}(\varphi)[v]_{\mathcal{B}}$).

Definition. The $m \times n$ matrix $A = (a_{ij})$ associated to the linear transformation φ above is said to *represent* the linear transformation φ with respect to the bases $\mathcal{B}, \mathcal{E}$. Similarly, φ is the linear transformation represented by A with respect to the bases $\mathcal{B}, \mathcal{E}$.

Examples

- (1) Let $V = \mathbb{R}^3$ with the standard basis $\mathcal{B} = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ and let $W = \mathbb{R}^2$ with the standard basis $\mathcal{E} = \{(1, 0), (0, 1)\}$. Let φ be the linear transformation $\varphi(x, y, z) = (x + 2y, x + y + z)$. Since $\varphi(1, 0, 0) = (1, 1)$, $\varphi(0, 1, 0) = (2, 1)$, $\varphi(0, 0, 1) = (0, 1)$, the matrix $A = M_{\mathcal{B}}^{\mathcal{E}}(\varphi)$ is the matrix $\begin{pmatrix} 1 & 2 & 0 \\ 1 & 1 & 1 \end{pmatrix}$.

- (2) Let $V = W$ be the 2-dimensional space of solutions of the differential equation $y'' - 3y' + 2y = 0$ over $\mathbb{C}$ and let $\mathcal{B} = \mathcal{E}$ be the basis $v_1 = e^t$, $v_2 = e^{2t}$. Since the coefficients of this equation are constants it is easy to check that if y is a solution then its derivative y' is also a solution. It follows that the map $\varphi = d/dt =$ differentiation (with respect to t) is a linear transformation from V to itself. Since $\varphi(v_1) = d(e^t)/dt = e^t = v_1$ and $\varphi(v_2) = d(e^{2t})/dt = 2e^{2t} = 2v_2$ we see that the corresponding matrix with respect to these bases is the diagonal matrix $\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$.
- (3) Let $V = W = \mathbb{Q}^3 = \{(x, y, z) \mid x, y, z \in \mathbb{Q}\}$ be the usual 3-dimensional vector space of ordered 3-tuples with entries from the field $F = \mathbb{Q}$ of rational numbers and suppose φ is the linear transformation

$$\varphi(x, y, z) = (9x + 4y + 5z, -4x - 3z, -6x - 4y - 2z), \quad x, y, z \in \mathbb{Q}$$

from V to itself. Take the standard basis $e_1 = (1, 0, 0)$, $e_2 = (0, 1, 0)$, $e_3 = (0, 0, 1)$ for V and for $W = V$. Since $\varphi(1, 0, 0) = (9, -4, -6)$, $\varphi(0, 1, 0) = (4, 0, -4)$, $\varphi(0, 0, 1) = (5, -3, -2)$, the matrix A representing this linear transformation with respect to these bases is

$$A = \begin{pmatrix} 9 & 4 & 5 \\ -4 & 0 & -3 \\ -6 & -4 & -2 \end{pmatrix}.$$

Theorem 10. Let V be a vector space over F of dimension n and let W be a vector space over F of dimension m , with bases $\mathcal{B}$, $\mathcal{E}$ respectively. Then the map $\text{Hom}_F(V, W) \rightarrow M_{m \times n}(F)$ from the space of linear transformations from V to W to the space of $m \times n$ matrices with coefficients in F defined by $\varphi \mapsto M_{\mathcal{B}}^{\mathcal{E}}(\varphi)$ is a vector space isomorphism. In particular, there is a bijective correspondence between linear transformations and their associated matrices with respect to a fixed choice of bases.

Proof: The columns of the matrix $M_{\mathcal{B}}^{\mathcal{E}}(\varphi)$ are determined by the action of φ on the basis $\mathcal{B}$ as in equation (3). This shows in particular that the map $\varphi \mapsto M_{\mathcal{B}}^{\mathcal{E}}(\varphi)$ is an F -linear map since φ is F -linear. This map is *surjective* since given a matrix M , the map φ defined by equation (3) on a basis and then extended by linearity is a linear transformation with matrix M . The map is *injective* since two linear transformations agreeing on a basis are the same.

Note that different choices of bases give rise to different isomorphisms, so in the same sense that there is no natural choice of basis for a vector space, there is no natural isomorphism between $\text{Hom}_F(V, W)$ and $M_{m \times n}(F)$.

Corollary 11. The dimension of $\text{Hom}_F(V, W)$ is $(\dim V)(\dim W)$.

Proof: The dimension of $M_{m \times n}(F)$ is mn .

Definition. An $m \times n$ matrix A is called *nonsingular* if $Ax = 0$ with $x \in F^n$ implies $x = 0$.

The connection of **the term nonsingular** applied to **matrices** and to linear transformations is the following: **let** $A = M_B^E(\varphi)$ be the matrix **associated** to the linear transformation φ (with some **choice of bases** B, E). Then **independently** of the choice of bases, the $m \times n$ matrix A is nonsingular **if and only if** the linear transformation φ is a nonsingular linear transformation from the n -dimensional space V **to the** m -dimensional space W (cf. the exercises).

Assume now that U, V and W are all finite dimensional vector spaces over F with ordered bases $\mathcal{D}, \mathcal{B}$ and $\mathcal{E}$ respectively, where $\mathcal{B}$ and $\mathcal{E}$ are as **before** and suppose $\mathcal{D} = \{u_1, u_2, \dots, u_k\}$. Assume $\psi : U \rightarrow V$ and $\varphi : V \rightarrow W$ are linear transformations. Their composite, $\varphi \circ \psi$, is a linear transformation from U to W , so we **can compute** its matrix with respect to the appropriate bases; namely, $M_D^E(\varphi \circ \psi)$ is found **by computing**

$$\varphi \circ \psi(u_j) = \sum_{i=1}^m \gamma_{ij} w_i$$

and putting the coefficients γ_{ij} down the j^{th} column of $M_D^E(\varphi \circ \psi)$. Next, **compute the** matrices of ψ and φ separately:

$$\psi(u_j) = \sum_{p=1}^n \alpha_{pj} v_p \quad \text{and} \quad \varphi(v_p) = \sum_{i=1}^m \beta_{ip} w_i$$

so that $M_D^B(\psi) = (\alpha_{pj})$ and $M_B^E(\varphi) = (\beta_{ip})$.

Using these coefficients we can find an expression for the γ 's in terms of **the** α 's and β 's as follows:

$$\begin{aligned} \varphi \circ \psi(u_j) &= \varphi\left(\sum_{p=1}^n \alpha_{pj} v_p\right) \\ &= \sum_{p=1}^n \alpha_{pj} \varphi(v_p) \\ &= \sum_{p=1}^n \alpha_{pj} \sum_{i=1}^m \beta_{ip} w_i \\ &= \sum_{p=1}^n \sum_{i=1}^m \alpha_{pj} \beta_{ip} w_i. \end{aligned}$$

By interchanging the order of summation in the above double sum we see that γ_{ij} , **which** is the coefficient of w_i in the above expression, is

$$\gamma_{ij} = \sum_{p=1}^n \alpha_{pj} \beta_{ip}.$$

Computing the product of the matrices for φ and ψ (in that order) we obtain

$$(\beta_{ij})(\alpha_{ij}) = (\delta_{ij}), \quad \text{where} \quad \delta_{ij} = \sum_{p=1}^m \beta_{ip} \alpha_{pj}.$$

By comparing the two sums above and using the commutativity of field multiplication, we see that for all i and j , $\gamma_{ij} = \delta_{ij}$. This computation proves the following result:

Theorem 12. With notations as above, $M_D^{\mathcal{E}}(\varphi \circ \psi) = M_B^{\mathcal{E}}(\varphi)M_D^{\mathcal{B}}(\psi)$, i.e., with respect to a compatible choice of bases, the product of the matrices representing the linear transformations φ and ψ is the matrix representing the composite linear transformation $\varphi \circ \psi$.

Corollary 13. Matrix multiplication is associative and distributive (whenever the dimensions are such as to make products defined). An $n \times n$ matrix A is nonsingular if and only if it is invertible.

Proof: Let A , B and C be matrices such that the products $(AB)C$ and $A(BC)$ are defined, and let S , T and R denote the associated linear transformations. By Theorem 12, the linear transformation corresponding to AB is the composite $S \circ T$ so the linear transformation corresponding to $(AB)C$ is the composite $(S \circ T) \circ R$. Similarly, the linear transformation corresponding to $A(BC)$ is the composite $S \circ (T \circ R)$. Since function composition is associative, these two linear transformations are the same, and so $(AB)C = A(BC)$ by Theorem 10. The distributivity is proved similarly. Note also that it is possible to prove these results by straightforward (albeit tedious) calculations with matrices.

If A is invertible, then $Ax = 0$ implies $x = A^{-1}Ax = A^{-1}0 = 0$, so A is nonsingular. Conversely, if A is nonsingular, fix bases $\mathcal{B}$, $\mathcal{E}$ for V and let φ be the linear transformation of V to itself represented by A with respect to these bases. By Corollary 9, φ is an isomorphism of V to itself, hence has an inverse, φ^{-1} . Let B be the matrix representing φ^{-1} with respect to the bases $\mathcal{E}$, $\mathcal{B}$ (note the order). Then $AB = M_B^{\mathcal{E}}(\varphi)M_E^{\mathcal{B}}(\varphi^{-1}) = M_E^{\mathcal{E}}(\varphi \circ \varphi^{-1}) = M_E^{\mathcal{E}}(1) = I$. Similarly, $BA = I$ so B is the inverse of A .

Corollary 14.

- (1) If $\mathcal{B}$ is a basis of the n -dimensional space V , the map $\varphi \mapsto M_B^{\mathcal{B}}(\varphi)$ is a ring and a vector space isomorphism of $\text{Hom}_F(V, V)$ onto the space $M_n(F)$ of $n \times n$ matrices with coefficients in F .
- (2) $GL(V) \cong GL_n(F)$ where $\dim V = n$. In particular, if F is a finite field the order of the finite group $GL_n(F)$ (which equals $|GL(V)|$) is given by the formula at the end of Section 1.

Proof: (1) We have already seen in Theorem 10 that this map is an isomorphism of vector spaces over F . Corollary 13 shows that $M_n(F)$ is a ring under matrix multiplication, and then Theorem 12 shows that multiplication is preserved under this map, hence it is also a ring isomorphism.

(2) This is immediate from (1) since a ring isomorphism sends units to units.

Definition. If A is any $m \times n$ matrix with entries from F , the *row rank* (respectively, *column rank*) of A is the maximal number of linearly independent rows (respectively,

columns) of A (where the rows or columns of A are considered as vectors in affine n -space, m -space, respectively).

The relation between the rank of a matrix and the rank of the associated linear transformation is the following: the rank of φ as a linear transformation equals the column rank of the matrix $M_{\mathcal{B}}^{\mathcal{E}}(\varphi)$ (cf. the exercises). We shall also see that the row rank and the column rank of any matrix are the same.

We now consider the relation of two matrices associated to the same linear transformation of a vector space to itself but with respect to two different choices of bases (cf. the exercises for the general statement regarding a linear transformation from a vector space V to another vector space W).

Definition. Two $n \times n$ matrices A and B are said to be *similar* if there is an invertible (i.e., nonsingular) $n \times n$ matrix P such that $P^{-1}AP = B$. Two linear transformations φ and ψ from a vector space V to itself are said to be *similar* if there is a nonsingular linear transformation ξ from V to V such that $\xi^{-1}\varphi\xi = \psi$.

Suppose $\mathcal{B}$ and $\mathcal{E}$ are two bases of the same vector space V and let $\varphi \in \text{Hom}_F(V, V)$. Let I be the identity map from V to V and let $P = M_{\mathcal{E}}^{\mathcal{B}}(I)$ be its associated matrix (in other words, write the elements of the basis $\mathcal{E}$ in terms of the basis $\mathcal{B}$ — note the order — and use the resulting coordinates for the columns of the matrix P). Note that if $\mathcal{B} \neq \mathcal{E}$ then P is *not* the identity matrix. Then $P^{-1}M_{\mathcal{B}}^{\mathcal{B}}(\varphi)P = M_{\mathcal{E}}^{\mathcal{E}}(\varphi)$. If $[v]_{\mathcal{B}}$ is the $n \times 1$ matrix of coordinates for $v \in V$ with respect to the basis $\mathcal{B}$, and similarly $[v]_{\mathcal{E}}$ is the $n \times 1$ matrix of coordinates for $v \in V$ with respect to the basis $\mathcal{E}$, then $[v]_{\mathcal{B}} = P[v]_{\mathcal{E}}$. The matrix P is called the *transition* or *change of basis* matrix from $\mathcal{B}$ to $\mathcal{E}$ and this similarity action on $M_{\mathcal{B}}^{\mathcal{B}}(\varphi)$ is called a *change of basis*. This shows that the matrices associated to the same linear transformation with respect to two different bases are similar.

Conversely, suppose A and B are $n \times n$ matrices similar by a nonsingular matrix P . Let $\mathcal{B}$ be a basis for the n -dimensional vector space V . Define the linear transformation φ of V (with basis $\mathcal{B}$) to V (again with basis $\mathcal{B}$) by equation (3) using the given matrix A , i.e.,

$$\varphi(v_j) = \sum_{i=1}^n \alpha_{ij} v_i.$$

Then $A = M_{\mathcal{B}}^{\mathcal{B}}(\varphi)$ by definition of φ . Define a new basis $\mathcal{E}$ of V by using the i^{th} column of P for the coordinates of w_i in terms of the basis $\mathcal{B}$ (so $P = M_{\mathcal{E}}^{\mathcal{B}}(I)$ by definition). Then $B = P^{-1}AP = P^{-1}M_{\mathcal{B}}^{\mathcal{B}}(\varphi)P = M_{\mathcal{E}}^{\mathcal{E}}(\varphi)$ is the matrix associated to φ with respect to the basis $\mathcal{E}$. This shows that any two similar $n \times n$ matrices arise in this fashion as the matrices representing the same linear transformation with respect to two different choices of bases.

Note that change of basis for a linear transformation from V to itself is the same as conjugation by some element of the group $GL(V)$ of nonsingular linear transformations of V to V . In particular, the relation “similarity” is an equivalence relation whose equivalence classes are the orbits of $GL(V)$ acting by conjugation on $\text{Hom}_F(V, V)$. If

$\varphi \in GL(V)$ (i.e., φ is an invertible linear transformation), then the similarity class of φ is none other than the conjugacy class of φ in the group $GL(V)$.

Example

Let $V = \mathbb{Q}^3$ and let φ be the linear transformation

$$\varphi(x, y, z) = (9x + 4y + 5z, -4x - 3z, -6x - 4y - 2z), \quad x, y, z \in \mathbb{Q}$$

from V to itself we considered in an earlier example. With respect to the standard basis, $\mathcal{B}$, $b_1 = (1, 0, 0)$, $b_2 = (0, 1, 0)$, $b_3 = (0, 0, 1)$ we saw that the matrix A representing this linear transformation is

$$A = M_{\mathcal{B}}^{\mathcal{B}}(\varphi) = \begin{pmatrix} 9 & 4 & 5 \\ -4 & 0 & -3 \\ -6 & -4 & -2 \end{pmatrix}.$$

Take now the basis, $\mathcal{E}$, $e_1 = (2, -1, -2)$, $e_2 = (1, 0, -1)$, $e_3 = (3, -2, -2)$ for V (we shall see that this is in fact a basis momentarily). Since

$$\varphi(e_1) = \varphi(2, -1, -2) = (4, -2, -4) = 2 \cdot e_1 + 0 \cdot e_2 + 0 \cdot e_3$$

$$\varphi(e_2) = \varphi(1, 0, -1) = (4, -1, -4) = 1 \cdot e_1 + 2 \cdot e_2 + 0 \cdot e_3$$

$$\varphi(e_3) = \varphi(3, -2, -2) = (9, -6, -6) = 0 \cdot e_1 + 0 \cdot e_2 + 3 \cdot e_3,$$

the matrix representing φ with respect to this basis is the matrix

$$B = M_{\mathcal{E}}^{\mathcal{E}}(\varphi) = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

Writing the elements of the basis $\mathcal{E}$ in terms of the basis $\mathcal{B}$ we have

$$e_1 = 2b_1 - b_2 - 2b_3$$

$$e_2 = b_1 - b_3$$

$$e_3 = 3b_1 - 2b_2 - 2b_3$$

so the matrix $P = M_{\mathcal{E}}^{\mathcal{B}}(I) = \begin{pmatrix} 2 & 1 & 3 \\ -1 & 0 & -2 \\ -2 & -1 & -2 \end{pmatrix}$ with inverse $P^{-1} = \begin{pmatrix} -2 & -1 & -2 \\ 2 & 2 & 1 \\ 1 & 0 & 1 \end{pmatrix}$

conjugates A into B , i.e., $P^{-1}AP = B$, as can easily be checked. (Note incidentally that since P is invertible this proves that $\mathcal{E}$ is indeed a basis for V .)

We observe in passing that the matrix B representing this linear transformation φ is much simpler than the matrix A representing φ . The study of the simplest possible matrix representing a given linear transformation (and which basis to choose to realize it) is the study of *canonical forms* considered in the next chapter.

Linear Transformations on Tensor Products of Vector Spaces

For convenience we reiterate Corollaries 18 and 19 of Section 10.4 for the special case of vector spaces.

Proposition 15. Let F be a subfield of the field K . If W is an m -dimensional vector space over F with basis $w_1, \dots, w_m$, then $K \otimes_F W$ is an m -dimensional vector space over K with basis $1 \otimes w_1, \dots, 1 \otimes w_m$.

Proposition 16. Let V and W be finite dimensional vector spaces over the field F with bases $v_1, \dots, v_n$ and $w_1, \dots, w_m$ respectively. Then $V \otimes_F W$ is a vector space over F of dimension nm with basis $v_i \otimes w_j$, $1 \leq i \leq n$ and $1 \leq j \leq m$.

Remark: If v and w are nonzero elements of V and W , respectively, then it follows from the proposition that $v \otimes w$ is a nonzero element of $V \otimes_F W$, because we may always build bases of V and W whose first basis vectors are v , w , respectively. In a tensor product $M \otimes_R N$ of two R -modules where R is not a field it is in general substantially more difficult to determine when the tensor product $m \otimes n$ of two nonzero elements is zero.

Now let V, W, X, Y be finite dimensional vector spaces over F and let

$$\varphi : V \rightarrow X \quad \text{and} \quad \psi : W \rightarrow Y$$

be linear transformations. We compute a matrix of the linear transformation

$$\varphi \otimes \psi : V \otimes W \rightarrow X \otimes Y.$$

Let $\mathcal{B}_1 = \{v_1, \dots, v_n\}$ and $\mathcal{B}_2 = \{w_1, \dots, w_m\}$ be (ordered) bases of V and W respectively, and let $\mathcal{E}_1 = \{x_1, \dots, x_r\}$ and $\mathcal{E}_2 = \{y_1, \dots, y_s\}$ be (ordered) bases of X and Y respectively. Let $\mathcal{B} = \{v_i \otimes w_j\}$ and $\mathcal{E} = \{x_i \otimes y_j\}$ be the bases of $V \otimes W$ and $X \otimes Y$ given by Proposition 16; we shall order these shortly. Suppose

$$\varphi(v_i) = \sum_{p=1}^r \alpha_{pi} x_p \quad \text{and} \quad \psi(w_j) = \sum_{q=1}^s \beta_{qj} y_q.$$

Then

$$\begin{aligned} (\varphi \otimes \psi)(v_i \otimes w_j) &= (\varphi(v_i)) \otimes (\psi(w_j)) \\ &= \left(\sum_{p=1}^r \alpha_{pi} x_p \right) \otimes \left(\sum_{q=1}^s \beta_{qj} y_q \right) \\ &= \sum_{p=1}^r \sum_{q=1}^s \alpha_{pi} \beta_{qj} (x_p \otimes y_q). \end{aligned} \tag{11.8}$$

In view of the order of summation in (11.8) we order the basis $\mathcal{E}$ into r ordered sets, with the p^{th} list being $x_p \otimes y_1, x_p \otimes y_2, \dots, x_p \otimes y_s$, and similarly order the basis $\mathcal{B}$. Then equation (8) determines the column entries for the corresponding matrix of $\varphi \otimes \psi$. The resulting matrix $M_{\mathcal{B}}^{\mathcal{E}}(\varphi \otimes \psi)$ is an $r \times n$ block matrix whose p, q block is the $s \times m$ matrix $\alpha_{p,q} M_{\mathcal{B}_2}^{\mathcal{E}_2}(\psi)$. In other words, the matrix for $\varphi \otimes \psi$ is obtained by taking the matrix for φ and multiplying each entry by the matrix for ψ . Such matrices have a name:

Definition. Let $A = (\alpha_{ij})$ and B be $r \times n$ and $s \times m$ matrices, respectively, with coefficients from any commutative ring. The *Kronecker product* or *tensor product* of A and B , denoted by $A \otimes B$, is the $rs \times nm$ matrix consisting of an $r \times n$ block matrix whose i, j block is the $s \times m$ matrix $\alpha_{ij} B$.

With this terminology we have

Proposition 17. Let $\varphi : V \rightarrow X$ and $\psi : W \rightarrow Y$ be linear transformations of finite dimensional vector spaces. Then the Kronecker product of matrices representing φ and ψ is a matrix representation of $\varphi \otimes \psi$.

Example

Let $V = X = \mathbb{R}^3$, both with basis v_1, v_2, v_3 , and $W = Y = \mathbb{R}^2$, both with basis w_1, w_2 . Suppose $\varphi : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is the linear transformation given by $\varphi(av_1 + bv_2 + cv_3) = cv_1 + 2av_2 - 3bv_3$ and $\psi : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is the linear transformation given by $\psi(aw_1 + bw_2) = (a + 3b)w_1 + (4b - 2a)w_2$. With respect to the chosen bases, the matrices for φ and ψ are

$$\begin{pmatrix} 0 & 0 & 1 \\ 2 & 0 & 0 \\ 0 & -3 & 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 3 \\ -2 & 4 \end{pmatrix},$$

respectively. Then with respect to the ordered basis

$$\mathcal{B} = \{v_1 \otimes w_1, v_1 \otimes w_2, v_2 \otimes w_1, v_2 \otimes w_2, v_3 \otimes w_1, v_3 \otimes w_2\}$$

we have

$$M_{\mathcal{B}}^{\mathcal{B}}(\varphi \otimes \psi) = \left(\begin{array}{cc|cc|cc} 0 & 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & -2 & 4 \\ \hline 2 & 6 & 0 & 0 & 0 & 0 \\ -4 & 8 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & -3 & -9 & 0 & 0 \\ 0 & 0 & 6 & -12 & 0 & 0 \end{array} \right),$$

obtained (as indicated by the dashed lines) by multiplying the 2×2 matrix for ψ successively by the entries in the matrix for φ .

EXERCISES

1. Let V be the collection of polynomials with coefficients in $\mathbb{Q}$ in the variable x of degree at most 5. Determine the transition matrix from the basis $1, x, x^2, \dots, x^5$ for V to the basis $1, 1+x, 1+x+x^2, \dots, 1+x+x^2+x^3+x^4+x^5$ for V .
2. Let V be the vector space of the preceding exercise. Let $\varphi = d/dx$ be the linear transformation of V to itself given by usual differentiation of a polynomial with respect to x . Determine the matrix of φ with respect to the two bases for V in the previous exercise.
3. Let V be the collection of polynomials with coefficients in F in the variable x of degree at most n . Determine the transition matrix from the basis $1, x, x^2, \dots, x^n$ for V to the elements

$$1, x - \lambda, \dots, (x - \lambda)^{n-1}, (x - \lambda)^n$$

where λ is a fixed element of F . Conclude that these elements are a basis for V .

4. Let φ be the linear transformation of $\mathbb{R}^2$ to itself given by rotation counterclockwise around the origin through an angle θ . Show that the matrix of φ with respect to the standard basis for $\mathbb{R}^2$ is $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$.
5. Show that the $m \times n$ matrix A is nonsingular if and only if the linear transformation φ is a nonsingular linear transformation from the n -dimensional space V to the m -dimensional space W , where $A = M_{\mathcal{B}}^{\mathcal{E}}(\varphi)$, regardless of the choice of bases $\mathcal{B}$ and $\mathcal{E}$.

6. Prove if $\varphi \in \text{Hom}_F(F^n, F^m)$, and B, E are the natural bases of F^n, F^m respectively, then the range of φ equals the span of the set of columns of $M_B^E(\varphi)$. Deduce that the rank of φ (as a linear transformation) equals the column rank of $M_B^E(\varphi)$.
7. Prove that any two similar matrices have the same row rank and the same column rank.
8. Let V be an n -dimensional vector space over F and let φ be a linear transformation of the vector space V to itself.
- Prove that if V has a basis consisting of eigenvectors for φ (cf. Exercise 8 of Section 1) then the matrix representing φ with respect to this basis (for both domain and range) is diagonal with the eigenvalues as diagonal entries.
 - If A is the $n \times n$ matrix representing φ with respect to a given basis for V (for both domain and range) prove that A is similar to a diagonal matrix if and only if V has a basis of eigenvectors for φ .
9. If W is a subspace of the vector space V stable under the linear transformation φ (i.e., $\varphi(W) \subseteq W$), show that φ induces linear transformations $\varphi|_W$ on W and $\tilde{\varphi}$ on the quotient vector space V/W . If $\varphi|_W$ and $\tilde{\varphi}$ are nonsingular prove φ is nonsingular. Prove the converse holds if V has finite dimension and give a counterexample with V infinite dimensional.
10. Let V be an n -dimensional vector space and let φ be a linear transformation of V to itself. Suppose W is a subspace of V of dimension m that is stable under φ .
- Prove that there is a basis for V with respect to which the matrix for φ is of the form

$$\begin{pmatrix} A & B \\ 0 & C \end{pmatrix}$$

where A is an $m \times m$ matrix, B is an $m \times (n-m)$ matrix and C is an $(n-m) \times (n-m)$ matrix (such a matrix is called *block upper triangular*).

- Prove that if there is a subspace W' invariant under φ so that $V = W \oplus W'$ decomposes as a direct sum then the bases for W and W' give a basis for V with respect to which the matrix for φ is *block diagonal*:

$$\begin{pmatrix} A & 0 \\ 0 & C \end{pmatrix}$$

where A is an $m \times m$ matrix and C is an $(n-m) \times (n-m)$ matrix.

- Prove conversely that if there is a basis for V with respect to which φ is block diagonal as in (b) then there are φ -invariant subspaces W and W' of dimensions m and $n-m$, respectively, with $V = W \oplus W'$.
11. Let φ be a linear transformation from the finite dimensional vector space V to itself such that $\varphi^2 = \varphi$.
- Prove that $\text{image } \varphi \cap \ker \varphi = 0$.
 - Prove that $V = \text{image } \varphi \oplus \ker \varphi$.
 - Prove that there is a basis of V such that the matrix of φ with respect to this basis is a diagonal matrix whose entries are all 0 or 1.

A linear transformation φ satisfying $\varphi^2 = \varphi$ is called an *idempotent* linear transformation. This exercise proves that idempotent linear transformations are simply projections onto some subspace.

12. Let $V = \mathbb{R}^2$, $v_1 = (1, 0)$, $v_2 = (0, 1)$, so that v_1, v_2 are a basis for V . Let φ be the linear transformation of V to itself whose matrix with respect to this basis is $\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$. Prove that if W is the subspace generated by v_1 then W is stable under the action of φ . Prove that there is no subspace W' invariant under φ so that $V = W \oplus W'$.

13. Let V be a vector space of dimension n and let W be a vector space of dimension m over a field F . Suppose A is the $m \times n$ matrix representing a linear transformation φ from V to W with respect to the bases $\mathcal{B}_1$ for V and $\mathcal{E}_1$ for W . Suppose similarly that B is the $m \times n$ matrix representing φ with respect to the bases $\mathcal{B}_2$ for V and $\mathcal{E}_2$ for W . Let $P = M_{\mathcal{B}_2}^{\mathcal{B}_1}(I)$ where I denotes the identity map from V to V , and let $Q = M_{\mathcal{E}_2}^{\mathcal{E}_1}(I)$ where I denotes the identity map from W to W . Prove that $Q^{-1} = M_{\mathcal{E}_1}^{\mathcal{E}_2}(I)$ and that $Q^{-1}AP = B$, giving the general relation between matrices representing the same linear transformation but with respect to different choices of bases.

The following exercises recall the *Gauss–Jordan* elimination process. This is one of the fastest computational methods for the solution of a number of problems involving vector spaces — solving systems of linear equations, determining inverses of matrices, computing determinants, determining the span of a set of vectors, determining linear independence of a set of vectors etc.

Consider the system of m linear equations

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= c_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= c_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= c_m \end{aligned} \tag{11.4}$$

in the n unknowns $x_1, x_2, \dots, x_n$ where $a_{ij}, c_i, i = 1, 2, \dots, m, j = 1, 2, \dots, n$ are elements of the field F . Associated to this system is the *coefficient matrix*:

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

and the *augmented matrix*:

$$(A | C) = \left(\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1n} & c_1 \\ a_{21} & a_{22} & \dots & a_{2n} & c_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & c_m \end{array} \right)$$

(the term *augmented* refers to the presence of the column matrix $C = (c_i)$ in addition to the coefficient matrix $A = (a_{ij})$). The set of solutions in F of this system of equations is not altered if we perform any of the following three operations:

- (1) interchange any two equations
- (2) add a multiple of one equation to another
- (3) multiply any equation by a nonzero element from F ,

which correspond to the following three *elementary row operations* on the augmented matrix:

- (1) interchange any two rows
- (2) add a multiple of one row to another
- (3) multiply any row by a unit in F , i.e., by any nonzero element in F .

If a matrix A can be transformed into a matrix C by a series of elementary row operations then A is said to be *row reduced* to C .

14. Prove that if A can be row reduced to C then C can be row reduced to A . Prove that the relation " $A \sim C$ if and only if A can be row reduced to C " is an equivalence relation. [Observe that the elementary row operations are reversible.]

Matrices lying in the same equivalence class under this equivalence relation are said to be *row equivalent*.

15. Prove that the row rank of two row equivalent matrices is the same. [It suffices to prove this for two matrices differing by an elementary row operation.]

An $m \times n$ matrix is said to be in *reduced row echelon form* if

- (a) the first nonzero entry a_{ij} in row i is 1 and all other entries in the corresponding j_i^{th} column are zero, and
 (b) $j_1 < j_2 < \dots < j_r$ where r is the number of nonzero rows, i.e., the number of initial zeros in each row is strictly increasing (hence the term *echelon*).

An augmented matrix $(A | C)$ is said to be in reduced row echelon form if its coefficient matrix A is in reduced row echelon form. For example, the following two matrices are in reduced row echelon form:

$$\left(\begin{array}{cccccc|c} 1 & 0 & 5 & 7 & 0 & 3 & 0 \\ 0 & 1 & -1 & 1 & 0 & -4 & -1 \\ 0 & 0 & 0 & 0 & 1 & 6 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right) \quad \left(\begin{array}{cccc|c} 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & -3 \end{array} \right)$$

(with $j_1 = 1$, $j_2 = 2$, $j_3 = 5$ for the first matrix and $j_1 = 2$, $j_2 = 4$ for the second matrix). The first nonzero entry in any given row of the coefficient matrix of a reduced row echelon augmented matrix (in position (i, j_i) by definition) is sometimes referred to as a *pivotal* element (so the pivotal elements in the first matrix are in positions (1,1), (2,2) and (3,5) and the pivotal elements in the second matrix are in positions (1,2) and (2,4)). The columns containing pivotal elements will be called *pivotal* columns and the columns of the coefficient matrix not containing pivotal elements will be called *nonpivotal*.

16. Prove by induction that any augmented matrix can be put in reduced row echelon form by a series of elementary row operations.
 17. Let A and C be two matrices in reduced row echelon form. Prove that if A and C are row equivalent then $A = C$.
 18. Prove that the row rank of a matrix in reduced row echelon form is the number of nonzero rows.
 19. Prove that the reduced row echelon forms of the matrices

$$\left(\begin{array}{cccccc|c} 1 & 1 & 4 & 8 & 0 & -1 & -1 \\ 1 & 2 & 3 & 9 & 0 & -5 & -2 \\ 0 & -2 & 2 & -2 & 1 & 14 & 3 \\ 1 & 4 & 1 & 11 & 0 & -13 & -4 \end{array} \right) \quad \left(\begin{array}{cccc|c} 0 & -3 & 3 & 1 & 5 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 2 & -2 & 0 & -3 \end{array} \right)$$

are the two matrices preceding Exercise 16.

The point of the reduced row echelon form is that the corresponding system of linear equations is in a particularly simple form, from which the solutions to the system $AX = C$ in (4) can be determined immediately:

20. (*Solving Systems of Linear Equations*) Let $(A' | C')$ be the reduced row echelon form of the augmented matrix $(A | C)$. The number of zero rows of A' is clearly at least as great as the number of zero rows of $(A' | C')$.

- (a) Prove that if the number of zero rows of A' is strictly larger than the number of zero rows of $(A' | C')$ then there are no solutions to $AX = C$.

By (a) we may assume that A' and $(A' | C')$ have the same number, r , of nonzero rows (so $n \geq r$).

- (b) Prove that if $r = n$ then there is precisely one solution to the system of equations $AX = C$.
- (c) Prove that if $r < n$ then there are infinitely many solutions to the system of equations $AX = C$. Prove in fact that the values of the $n - r$ variables corresponding to the nonpivotal columns of $(A' | C')$ can be chosen arbitrarily and that the remaining r variables corresponding to the pivotal columns of $(A' | C')$ are then determined uniquely.

21. Determine the solutions of the following systems of equations:

(a)

$$\begin{aligned} -3x + 3y + z &= 5 \\ x - y &= 0 \\ 2x - 2y &= -3 \end{aligned}$$

(b)

$$\begin{aligned} x - 2y + z &= 5 \\ x - 4y + 6z &= 10 \\ 4x - 11y + 11z &= 12 \end{aligned}$$

(c)

$$\begin{aligned} x - 2y + z &= 5 \\ y - 2z &= 17 \\ 2x - 3y &= 27 \end{aligned}$$

(d)

$$\begin{aligned} x + y - 3z + 2u &= 2 \\ 3x - 2y + 5z + u &= 1 \\ 6x + y - 4z + 3u &= 7 \\ 2x + 2y - 6z &= 4 \end{aligned}$$

(e)

$$\begin{aligned} x + y + 4z + 8u - w &= -1 \\ x + 2y + 3z + 9u - 5w &= -2 \\ -2y + 2z - 2u + v + 14w &= 3 \\ x + 4y + z + 11u - 13w &= -4 \end{aligned}$$

22. Suppose A and B are two row equivalent $m \times n$ matrices.

(a) Prove that the set

$$X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

of solutions to the homogeneous linear equations $AX = 0$ as in equation (4) above are the same as the set of solutions to the homogeneous linear equations $BX = 0$. [It suffices to prove this for two matrices differing by an elementary row operation.]

- (b) Prove that any linear dependence relation satisfied by the columns of A viewed as vectors in F^m is also satisfied by the columns of B .